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Ovation™ Digital Twin User Guide

EMERSON

About this manual

Welcome to the Emerson Ovation Digital Twin system. Ovation Digital Twin technology integrates simulation and plant control on a single platform, both running in parallel to enable real-time advanced testing of new operating strategies in a risk-free environment before changes are applied to the operating plant.

This manual provides an overview of the Ovation Digital Twin system, which is a software program that provides scalable engineered solutions for operator training and certification programs, procedure development and validation, control logic testing and verification, as well as engineering test bed scenarios for plant process improvements.

The information in this manual represents the recommended standards and procedures. If your system requires a different configuration, contact your Emerson service representative or sales office for help with the review of your system. All installation and maintenance procedures described in this document are performed by qualified personnel and that the equipment used is only for the purposes described. Using alternate methods of installation or configuration could yield undesirable results.

Conventions used in this manual

For security purposes, actual IP addresses are not used in Ovation user manuals. The IP addresses used in this manual are for example purposes only and should not be used in an actual system.

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Revision history

Version	Date	Changes	Applicable Releases
6	August 2025	• Updated the section *Ovation systems supported* (page 13) and *Operating system requirements* (page 13) to include Ovation 4.0 support. • Added the section *Ovation Digital Twin Control Builder Unit Converter* (page 73).	• Ovation Digital Twin release 3.09 and later systems. • Ovation 3.7 and later systems.
5	June 2025	• Updated the section *Operating system requirements (page 13)* to include information about the supported operating systems. • Updated the section *Other requirements (page 14)* to include information about the unsupported controller configurations. • Updated the section *Hardware requirements (page 14)* to include information about the virtualization technology. • Updated the section *Ovation Digital Twin Network Viewer (page 53)* to include information about the Find Results tab. • Updated the section *Ovation Digital Twin Config (page 44)* to include information about the Performance, Live Digital Twin, and Smart Grid Extensions tabs. • Updated the section *Ovation Digital Twin Advanced Edit (page 57)* to include information about the Graph Digitizer, Notes tab, and Add Attachment box. • Updated the section *Ovation Digital Twin Advanced Tuning (page 65)* to add detailed information about the Advanced Tuning window. • Updated the section *Ovation Digital Twin Media Tables (page 69)* to add detailed information about the Advanced Mode calculation. • Added the section *Network Time Protocol (NTP) (page 77)*. • Made miscellaneous changes and clarifications.	• Ovation Digital Twin release 3.08 and later systems. • Ovation 3.7 and later systems.
4	January 2024	• Updated the section *Hardware requirements (page 14)*. • Added the section *License requirements for the Ovation Digital Twin (page 14)*. • Added the section *Recommended methods for registering and renewing Ovation Digital Twin licenses (page 15)*. • Updated the section *Ovation Digital Twin algorithm suites available (page 19)*. • Updated the section *Installing the Ovation Digital Twin software (page 21)*. • Updated the section *What are Ovation Digital Twin Included Applications? (page 41)*. • Added the section *Troubleshooting (page 77)*. • Made miscellaneous changes and clarifications.	• Ovation Digital Twin release 3.0 and later systems. • Ovation 3.6 and later systems.
3	August 2022	• Updated product name to *Ovation Digital Twin* from the former *Ovation Embedded Simulation*. • Made miscellaneous changes and clarifications.	• Ovation Digital Twin release 2.83 and later systems. • Ovation 3.6 and later systems.

Version	Date	Changes	Applicable Releases
2	February 2021	• Added description of *Ovation Spreadsheet Tool (page 11)*. • Updated information for *Ovation 3.8 support and system requirements (page 13)*. • Updated information for the *OES Tray Application (page 42)*. • Updated information for the *OES Algorithm Status application (page 48)*. • Updated information for the *OES Media Tables application (page 69)*. • Made miscellaneous changes and clarifications.	• Ovation Embedded Simulation release 2.83 and later systems. • Ovation 3.6 and later systems.
1	April 2020	Initial release of this manual.	• Ovation Embedded Simulation release 2.76 and later systems. • Ovation 3.6 and later systems.

Contents

1 Introduction to Ovation Digital Twin

Topics covered in this section:

1.1 What is the Ovation Digital Twin solution?

Ovation Digital Twin provides a scalable engineered solution for operator training and certification programs, procedure development and validation, control logic testing and verification, as well as engineering test bed scenarios for plant process improvements.

Ovation Digital Twin provides a single, all-inclusive platform that uses already-familiar embedded control system-based models for simulation and control operations, as well as configuration and maintenance activities, and eliminates the need for specialized programming knowledge. The use of a common platform across both the plant DCS and training systems provides significant user benefits not currently available with other high-fidelity simulation platforms that employ third-party models.

Ovation Digital Twin uses the same platform for plant control and simulation, eliminating the need to interface with third-party models. High fidelity simulation algorithms based on dynamic, first-principle engineering and thermodynamic relationships are created with standard Ovation control system configuration and management software applications to provide a single, cohesive environment for plant control and simulation.

Ovation Digital Twin provides numerous benefits, including:

- Reduces total cost of ownership.
- Provides staff familiarity with common hardware and application tools, eliminating the need for specialized programming knowledge.
- Supports the unique ability to mix fidelities to meet training needs, budgets, and schedules.
- Provides flexibility to expand functionality with time.
- Provides online and offline simulation process model visibility, enabling simple edits/ modifications to be done by the customer, saving service costs and time.
- Simplifies maintenance by easily allowing the simulator to keep pace with control system changes.

1.2 What are the Ovation Digital Twin algorithms?

DCS-based simulation models, created through the use of advanced process algorithms that accurately reflect the operation and interaction of plant equipment, are "embedded" within a standard DCS platform. Algorithms are mathematical formulas that define the behavior of specific plant processes and equipment. The algorithm reads values (inputs) and writes values (outputs) into points to accomplish certain desired actions in the system.

The Ovation Control Builder creates control sheets for running both the live plant and the simulated plant. You place the Ovation Digital Twin algorithms on control sheets to tell the Ovation control system which algorithms to use, which points to associate with the algorithms, and in what order the algorithms should execute. The Control Builder combines multiple algorithms and control sheets to create an entire simulated control strategy for a system process.

Note

This manual assumes that you have a working knowledge of Ovation systems, including using the Control Builder. For more information, refer to the *Ovation Control Builder User Guide*.

1.3 What other Ovation products are used with Ovation Digital Twin?

Ovation Digital Twin with embedded Ovation algorithms uses the same hardware, operating system, and software application components of the accompanying Ovation plant control system. Use of a common platform across your control and training systems eliminates the need to learn to operate a new system of workstations, equipment, maintenance, and control features.

Ovation algorithms are built using standard Ovation Developer Studio and Control Builder engineering tools. The Ovation-based models run directly in the Ovation Virtual Controllers. Virtual Controllers replicate actual plant DCS Controllers and are used to execute simulation model operation and interactions. The Virtual Controllers are equipped with a flow-network solver that calculates the thermodynamics of multiphase plant processes and conservation equations. The solver provides stable pressure and flow responses to accurately reflect the operation and interaction of a plant's equipment during normal and abnormal operating conditions.

Control sheets used in the simulation models that represent plant subsystems are grouped together and loaded into Virtual Controllers. After the simulation control sheets have been saved, they can be viewed online at any Operator Station using Signal Diagrams.

Signal Diagrams provide the means to monitor and tune a control or simulation process. Signal diagrams can be opened in floating or dockable windows to view runtime status, navigate between control logic, process models, and input/output (I/O) signals, view and change algorithm details, and display tracking and I/O details.

You can use the Ovation Process Historian (OPH) to better understand the typical and abnormal behavior of your plant processes, to identify common trends, to explore abnormalities, and to diagnose process flaws and failures. The Historian can be used to collect process values and messages that are generated by your Ovation control system. The historian stores these values and messages in an optimized historical data store that runs on a Microsoft Windows platform. You can view and filter this information, or output it to your printers, files, e-mail, or Web pages. The historian can archive this information to a removable media or to a nearby or remote fixed disk storage unit.

In addition, trends provide additional insight into plant operations by showing the periodic changes between control signals, process values, and soft control points. Unlike other simulation systems that use different control system point names than those used within a simulator, Ovation Digital Twin uses simulation point names that are identical to the control system point names, with the addition of a suffix to identify a simulation point. Employing a common naming convention enhances the connectivity between the control logic and process models, simplifying the process of updating the simulator to match plant changes. It also enables the use of other standard DCS applications that the operator would be familiar with on the control system, such as the Operator Station Point Menu, Point Information, Point Review, Operator Event Messaging, Trends, and the Alarm Management system.

Note

For information on these Ovation products, refer to the *Ovation Control Builder User Guide*, the *Ovation Developer Studio User Guide*, the *Ovation Operator Station User Guide*, the *Ovation Process Historian User Guide*, the *Ovation Virtual Controller User Guide*, or contact your Emerson representative.

1.3.1 Ovation Digital Twin Standalone Tools

Several tools are available to work alongside the Ovation Digital Twin system:

- **Ovation Digital Twin Instructor Station** - The Ovation Digital Twin Instructor Station provides development and operator training tools for the Ovation Digital Twin software. This easy-to-use Windows-based application allows trainers to invoke malfunction scenarios and prior system states for operator drills and testing. The Ovation Digital Twin Instructor Station can also be used to control the Ovation simulator during the engineering phase of a project.
- **Ovation Digital Twin Training Kiosk** - The Ovation Digital Twin Training Kiosk provides trainees with a simple, full-screen user interface to run pre-defined training exercises at any time, without the need for an instructor. It offers a guided, distraction-free learning experience, where trainees follow step-by-step instructions to complete each exercise. Only the necessary training tools are visible, while access to other programs is restricted. Exercises end automatically with a summary report and can be completed individually or as part of a team.
- **Ovation Digital Twin Spreadsheet Tool** - The Ovation Digital Twin Spreadsheet Tool is a stand-alone application that supports advanced tuning, charting, and analysis of live Ovation data within a spreadsheet environment. The Ovation Digital Twin Spreadsheet Tool looks and feels similar to commonly available commercial spreadsheet software.

Note

For information on these tools, refer to the *Ovation Digital Twin Instructor Station User Guide*, the *Ovation Digital Twin Tools User Guide*, or contact your Emerson representative.

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2 System Requirements and Licensing

Topics covered in this section:

2.1 Ovation systems supported

The Ovation Digital Twin system supports the following Ovation system versions:

- Ovation 3.7.0
- Ovation 3.7.0 ESP
- Ovation 3.8.0
- Ovation 3.8.0 ESP
- Ovation 4.0

2.2 Operating system requirements

Depending upon the Ovation system you are running, the Ovation Digital Twin system can run on various Microsoft Windows operating systems. The Ovation Digital Twin supports the following versions of Windows:

Table 1. Supported operating systems

	Ovation 3.7.0	Ovation 3.7.0 ESP	Ovation 3.8.0	Ovation 3.8.0 ESP	Ovation 4.0
Windows Server 2016 64-bit	Supported	Supported	Supported	-	-
Windows Server 2019	-	Supported	Supported	-	-
Windows Server 2022	-	-	-	Supported	-
Windows Server 2025	-	-	-	-	Supported

Note

If you attempt to install Ovation Digital Twin in a version older than Ovation 3.7.0, you will be prompted to upgrade to Ovation 3.7.0 before proceeding.

2.3 Other requirements

- All necessary Ovation patches.
- All most recent Ovation Digital Twin patches. For more information, see the Ovation Digital Twin Required Patches document.
- Any applicable Windows patches.
- Access to Ovation network. IPv6: disabled.
- Ovation Digital Twin supported virtual controllers are: Virtual Controller (OCR400, OCR1100, OCR3000) and Virtual Compact Controller (OCC100).
- Model Server drop, Database Server, and other non-Virtual Controller Host machines set to Sim Operator Station Drop Type in Developer Studio.
- User performing installation must use domain-wide login (not local).

2.4 Hardware requirements

- Minimum of 5GB disk space on each machine.
- Model Server machine requires a significant CPU power (depending on the complexity and scale of the Ovation Digital Twin solution designed). Contact the Ovation Digital Twin support team for guidance when setting up Model Server hardware.
- All power-saving options must be disabled (turned off).
- All of the machines (especially the Model Server) must be running in high performance mode (both on hardware and software level).

Note

The Model Server machine should not be located on a system which is running as a Virtual Controller Host. Having both the Model Server and the Virtual Controller in the same system impairs performance.

Note

Emerson's recommended virtualization technology is Microsoft Hyper-V.

2.5 License requirements for Ovation Digital Twin

Note

The license registration procedure of the Ovation Digital Twin can be done only on the Ovation Database Server machine.

Licenses are stored on a License Server and are accessed through the Ovation Developer Studio.

To run Ovation Digital Twin, you must obtain the following licenses:

- Solver license
- Algorithm suite license
- Ovation Digital Twin Instructor Station license
- Ovation Digital Twin Tools license (if required)

To obtain license keys, you must record your system's locking code. The locking code can be accessed either in the License Manager of the Ovation Developer Studio or in the License Information window of the following:

- Ovation Digital Twin Instructor Station
- Ovation Digital Twin Tray App
- Ovation Digital Twin LogViewer

After accessing and recording the locking code, contact your Emerson representative during normal work hours. Depending on your situation, this contact might be your project engineer, after-market representative, or field service engineer. Contact Emerson prior to installations or upgrades to ensure the availability of required key codes or licenses. After appropriate license keys are obtained, enter them (page 15) into either the License Manager of Developer Studio, the License Information window of the Instructor Station, or the License Information window (page 17) of the Ovation Digital Twin.

Note

For more information about using Instructor Station, refer to the *Ovation Digital Twin Instructor Station User Guide*.

For more information about using Ovation Digital Twin Tools, refer to the *Ovation Digital Twin Tools User Guide*.

2.6 Recommended methods for registering and renewing Ovation Digital Twin licenses

There are several methods to register new or renewed licenses for Ovation Digital Twin components.

- Initial installation: Enter the license information during the initial Ovation Digital Twin software installation (page 21).

  Note

  Emerson recommends that for new installations, you register your licenses when prompted during the initial installation process.

- Registering or renewing licenses on existing Ovation Digital Twin installations: Emerson always recommends that you enter the licenses using an Ovation Developer Studio or Ovation Digital Twin Instructor Station located on a Database Server. If these methods are not available, you may also use the License Information window of the Ovation Digital Twin Tray App or the Ovation Digital Twin LogViewer to enter a renewal license key.

Note

For more information about using Ovation Developer Studio, refer to the *Ovation Developer Studio Users Guide*.

2.6.1 To register or renew the Ovation Digital Twin license using the Ovation Digital Twin Tray App or the Ovation Digital Twin LogViewer

Note

Emerson recommends that you register Ovation Digital Twin license using an Ovation Developer Studio or Ovation Digital Twin Instructor Station that is located on an Ovation Database Server machine. For more information about using Ovation Developer's Studio, refer to the *Ovation Developer Studio User Guide*. For more information about using Instructor Station, refer to the *Ovation Digital Twin Instructor Station User Guide*.

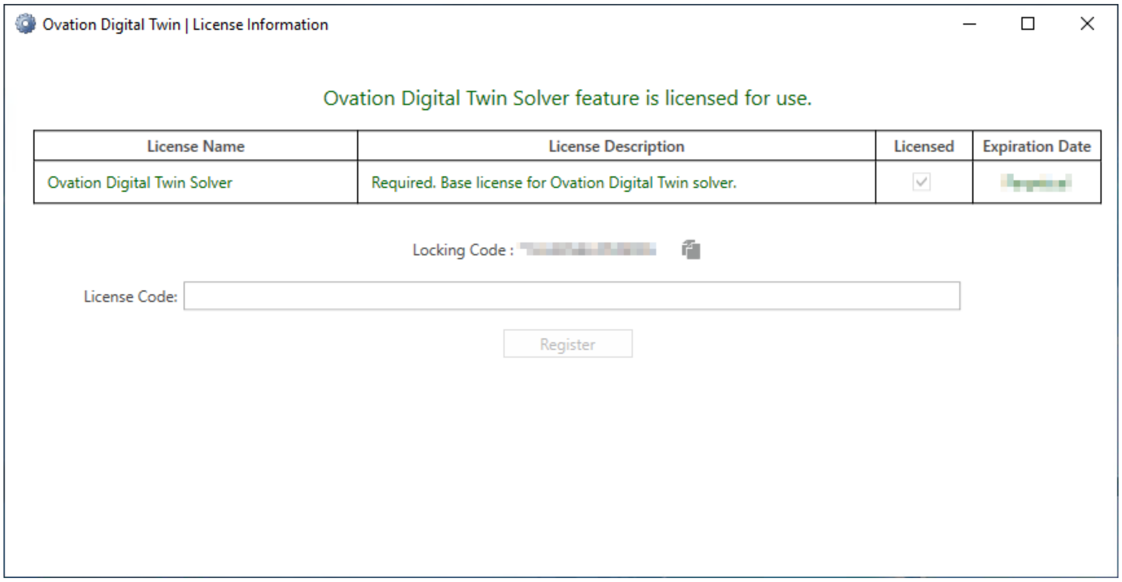
The license registration procedure of the Ovation Digital Twin can be done only on the Ovation Database Server machine.

If you have obtained new license keys from your Emerson representative and you do not have access to the Ovation Developer Studio or Ovation Digital Twin Instructor Station, use the following procedure to register or renew the Ovation Digital Twin solver license using the Ovation Digital Twin Tray App or LogViewer:

1. Access the Ovation Digital Twin Tray App (page 42) and click **License Status**.
   If no icon appears on the License Status option, your licensing is valid, and no further action is needed. If a warning or an error icon appears on the License Status option, continue with the procedure. For more information on the license status icons, see *License Information window* (page 17).
   OR
   Access the Ovation Digital Twin LogViewer (page 52) and click **License Information**.
   If the License Information button displays in green, your license is valid, and no further action is needed. If the button displays in yellow or red, continue with the procedure. For more information on the license status colors, see *License Information window* (page 17).

2. The License Information window appears. Enter the license code in the License Code field and click **Register**.

   Figure 1. License Information window

   If the registration is successful, a success message appears.
   If you attempt to register with an expired or invalid license, the registration fails, and an error message appears.

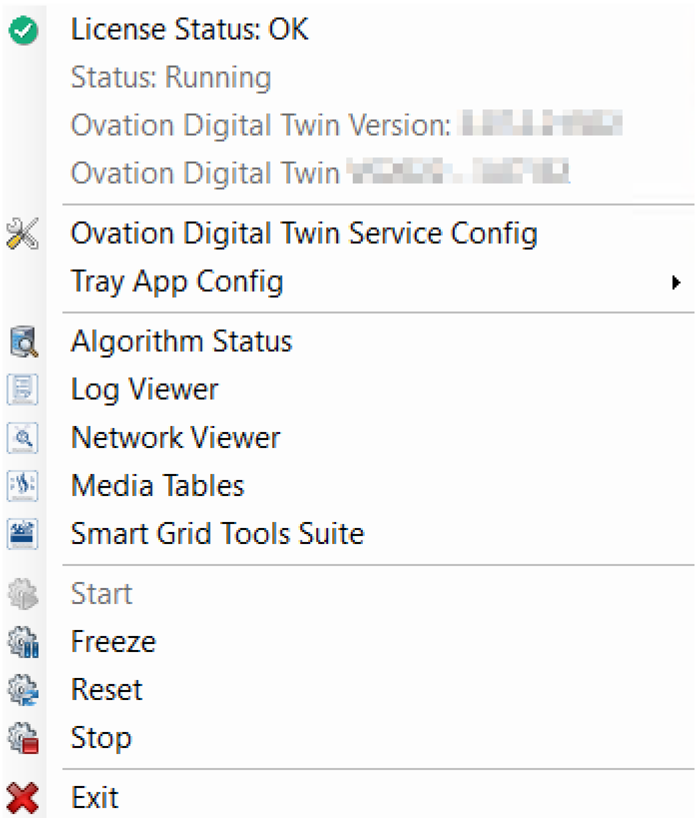
3. If you are registering a renewed license after the expiration of your previous license, you must restart your Ovation Digital Twin service. However, if you are registering a renewed license while the previous license is still valid, no further action is necessary.

2.6.2 License Information window

The License Information window provides the license name, description, license status, and any expiration date associated with the license. It also provides the locking code that you must have in order to obtain license keys.

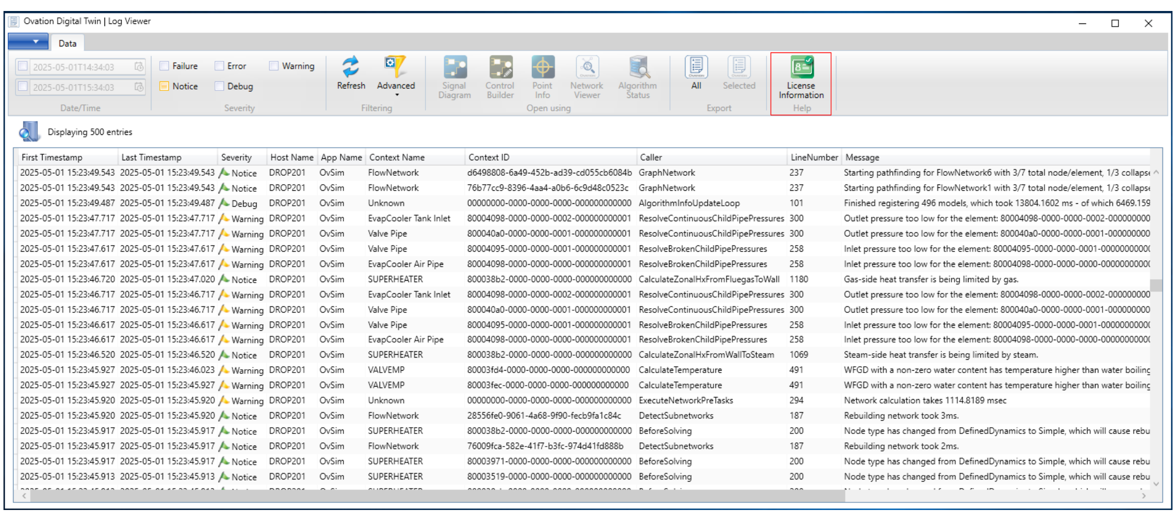
The License Information window can be accessed through either the License Status menu option of the Ovation Digital Twin Tray App or the License Information button of the Ovation Digital Twin LogViewer. The License Information window can also be accessed through the Instructor Station. For more information about using Instructor Station, refer to the *Ovation Digital Twin Instructor Station User Guide*.

Figure 2. Ovation Digital Twin Tray App control menu



The icon displayed on the License Status option varies depending on the current status of your license. A warning icon indicates that the license is set to expire within 90 days, and an error icon indicates that the license is expired.

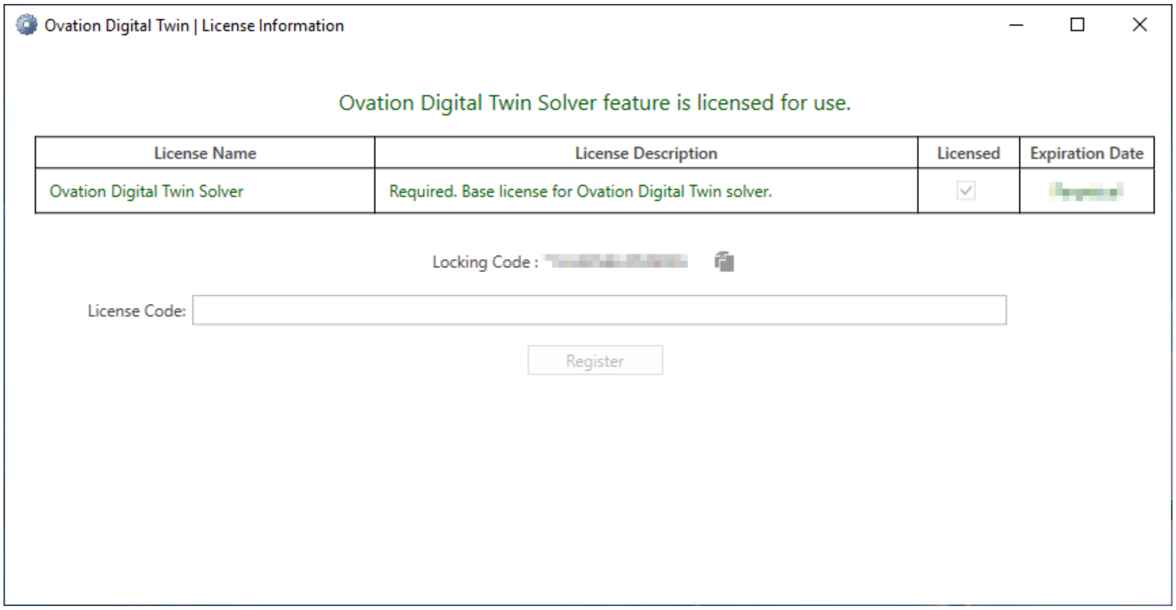
Figure 3. Ovation Digital Twin LogViewer - License Information icon



The License Information icon and the wording on the License Information window use different colors to indicate the current status of your license. The following License information colors are used:

- Green - The license is valid and does not expire within 90 days.
- Yellow - The license is set to expire within 90 days.
- Red - The license is expired.

Figure 4. License Information window (green font color indicating that the license is valid and does not expire within 90 days)



2.7 Ovation Digital Twin algorithm suites available

Ovation Digital Twin algorithms are licensed and grouped according to their function. There are 13 Ovation Digital Twin algorithm suites:

- Base algorithm suite
- Balance of Plant (BoP) algorithm suite
- Boiler algorithm suite
- Combined Cycle algorithm suite
- Electrical algorithm suite
- Flue Gas Desulfurization algorithm suite
- Internal Combustion Engine suite
- Live Digital Twin suite
- Selective Catalytic Reduction algorithm suite
- Turbine algorithm suite
- Hydropower suite
- Water Management suite
- Smart Grid Extensions suite

Note

For a list of Ovation Digital Twin algorithms currently included in each suite, consult the individual Ovation Digital Twin algorithm suite manuals or ask your Emerson representative.

3 Installing the Ovation Digital Twin software

Topics covered in this section:

3.1 Ovation Digital Twin installation overview

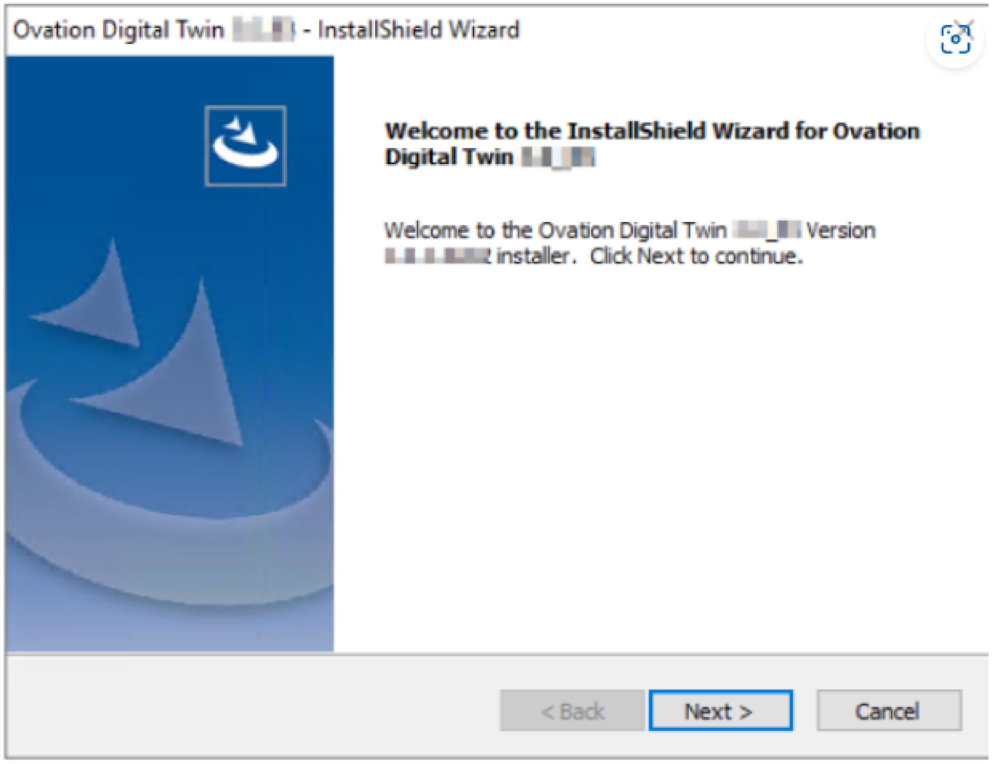
The following sections discuss installing Ovation Digital Twin software. The software must be installed on the following drops and in the following order:

1. Database Server
2. All Operator Stations (including Virtual Controller Host (VCH))
3. Model Server

3.1.1 To install the Ovation Digital Twin software

1. Insert the Ovation Digital Twin DVD into the DVD drive. If the program does not start automatically, open a Windows Explorer window, access the drive, and click **setup.exe**.
   The InstallShield Wizard page appears. Click **Next**.

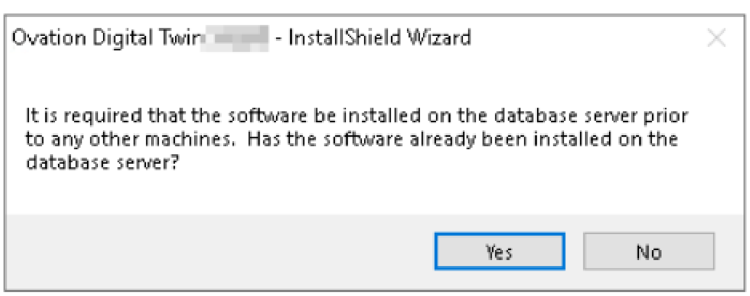
   Figure 5. InstallShield Wizard - Welcome page

   The installer validates your current operating system and Ovation installation. If all necessary components are installed, you can continue. If not, the installation aborts.

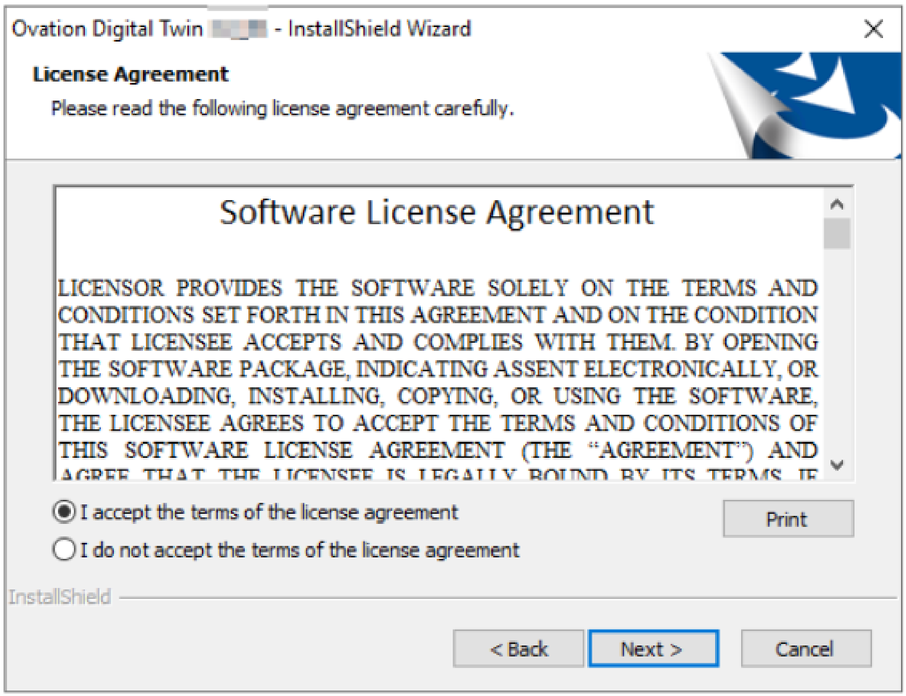
2. If the system does not recognize the drop as a Database Server, a message prompt appears, asking whether software has already been installed on the database server. Answer **Yes** or **No**.

   Figure 6. Database Server message prompt

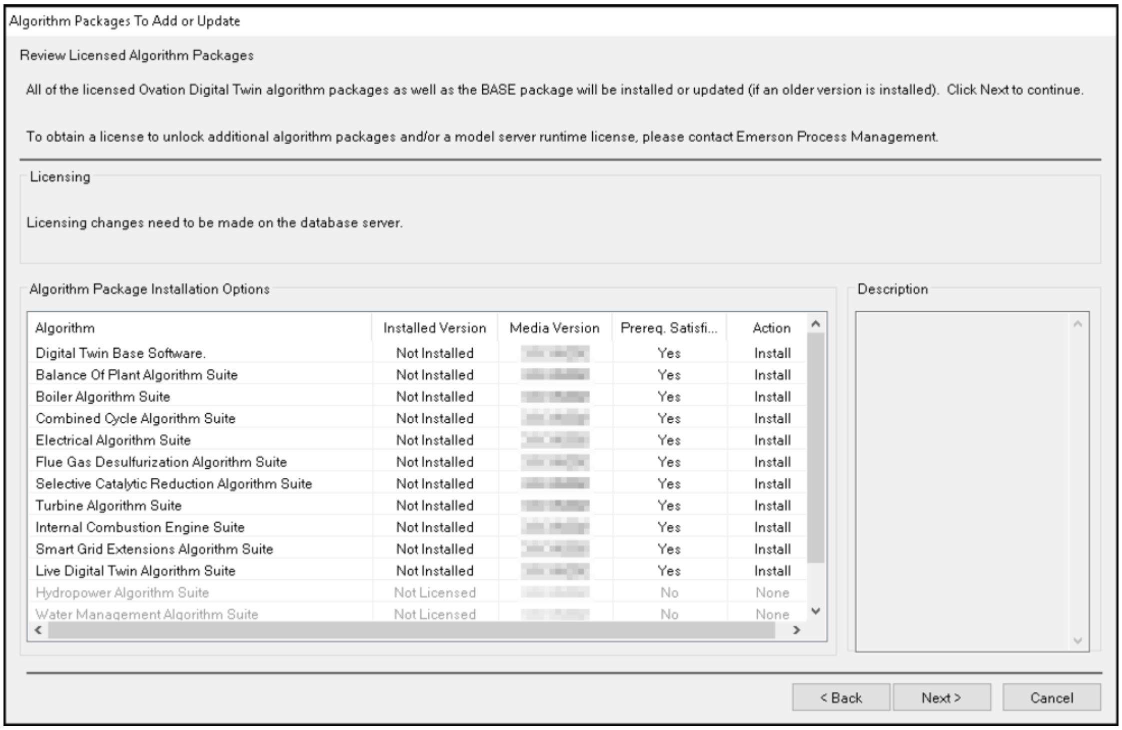
3. The License Agreement page appears. Read the terms and conditions of the license agreement. If you wish to print a copy of the agreement, click **Print**. To accept the terms and conditions of the agreement and continue with the installation, click **I accept the terms of the license agreement**, then click **Next.** Optionally, you may also click **Back** to return to the previous window or click **Cancel** to cancel the installation.

Figure 7. License Agreement page



4. After you accept the agreement and click **Next**, the Algorithm Packages to Add or Update window appears.

Figure 8. Algorithm Packages to Add or Update window



Note

If this program is installed on a Database Server, the licensing fields also appear on the window.

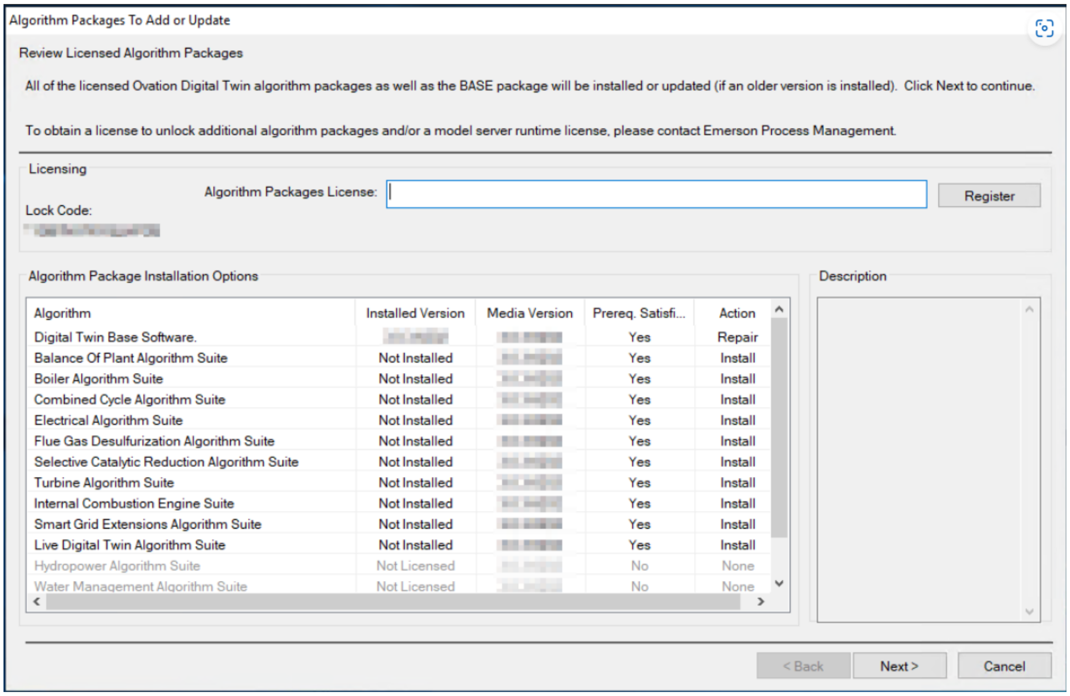
Perform the following:

a. Call Emerson's Licensing department and provide the **Lock Code** that appears on the window.

b. The Licensing department representative will give you an **Algorithm Packages License** key that contains the licenses for the Ovation Digital Twin algorithm suites that you obtained. Type the **license key** in the Algorithm Packages License field, then click **Register**. The Ovation Digital Twin algorithms that you have licensed are no longer grayed out.

c. The Licensing department representative will also provide a Model Server Runtime License key that contains the license for the model server. Type the **license key** in the Model Server Runtime License field, then click **Register**. The Model Server Runtime License key field disappears.

Figure 9. Algorithm Packages to Add or Update window (after Algorithm Packages License key is entered, showing selected, licensed algorithms)



d. Click **Next**.

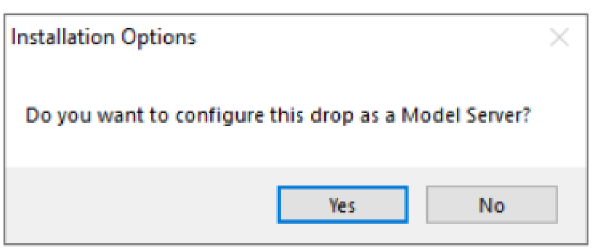
Note

When installing Ovation Digital Twin algorithms on a non-Database Server drop, the licensing parameter fields do not appear.

5. The Installation Options confirmation box appears, prompting you to confirm whether you want to configure this drop as a Model Server.

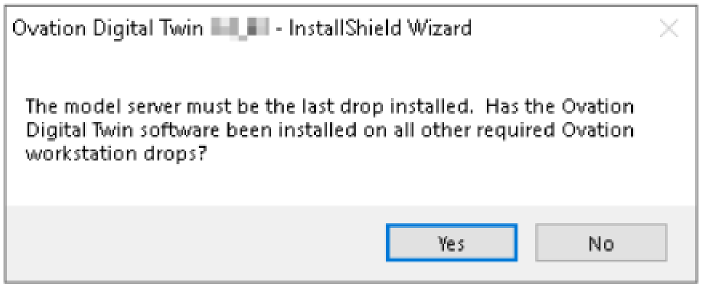
   - If you are installing a Database Server or other non-Model Server drop, then click **No** and proceed with Step 8.
   - If you are installing a Model Server drop, then click **Yes** and proceed with Step 6.

Figure 10. Installation Options window



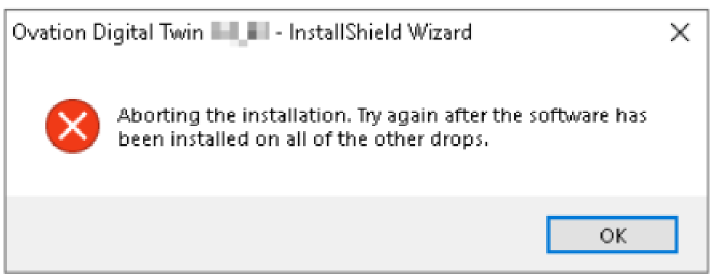
6. A warning message appears, prompting you to confirm whether the Ovation Digital Twin software has been installed on all other required Ovation workstation drops.
   - To confirm that the Ovation Digital Twin software has been installed on all other required Ovation workstation drops, click **Yes** and proceed with Step 8.
   - If you have not installed the Database Server and all other drops, then click **No** and proceed with Step 7.

Figure 11. Model Server warning prompt



7. A warning message appears stating that the installation will be aborted, as shown in the following figure. Click **OK** to exit the installation. Complete the installation for all other drops, then install the model server.

Figure 12. Message prompt aborting installation

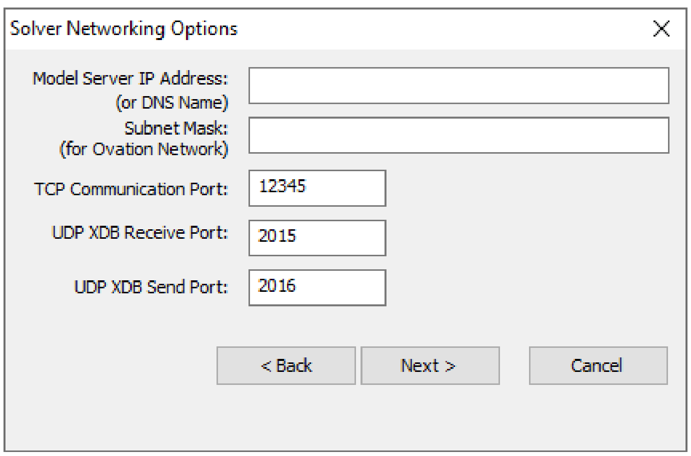


8. The Solver Networking Options window appears. Depending on the type of drop you are installing, some fields may be pre-populated with defaults but are still editable. If the fields are not pre-populated, complete them as follows:

   - **Model Server IP Address (or DNS name)** - Ovation network IP address of the model server. This field is initially blank on the Database Server and must be entered.
   - **Subnet Mask (for Ovation Network)** - Subnetwork mask of the Ovation network. To find it, use the following command from a command prompt:
     ipconfig/all.
   - **TCP Communication Port** - This value must be the same on all drops. The default value is 12345.
   - **UDP XDB Receive Port** - This value must be the same on all drops. The default value is 2015.
   - **UDP XDB Send Port** - This value must be the same on all drops. The default value is 2016.

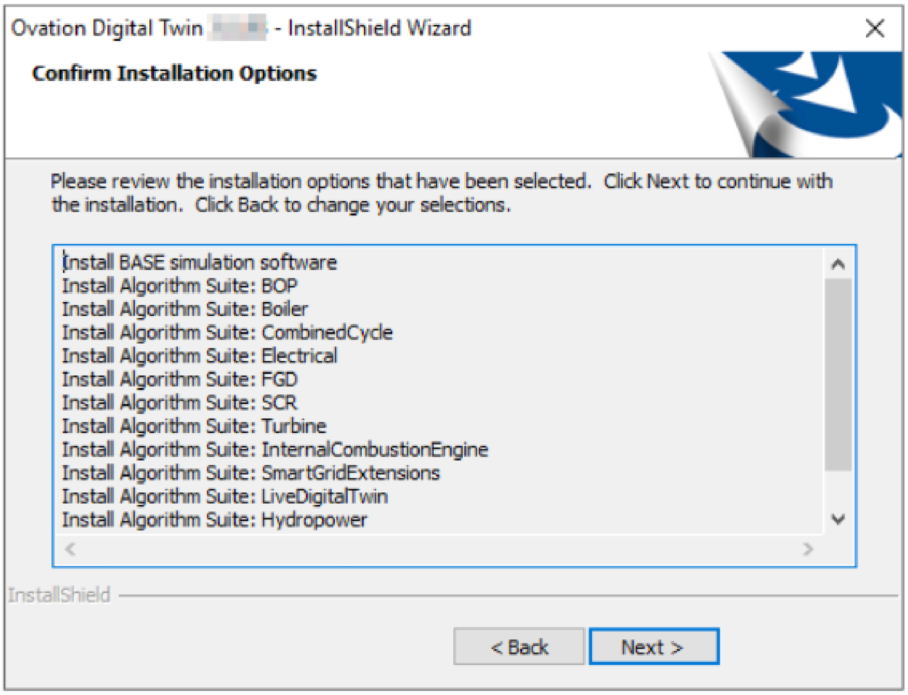
   When all fields are complete, click **Next.**

   Figure 13. Solver Networking Options window



9. The Confirm Installation Options page appears. Review the list of options that will be installed, then click **Next**.

Figure 14. Confirm Installation Options page



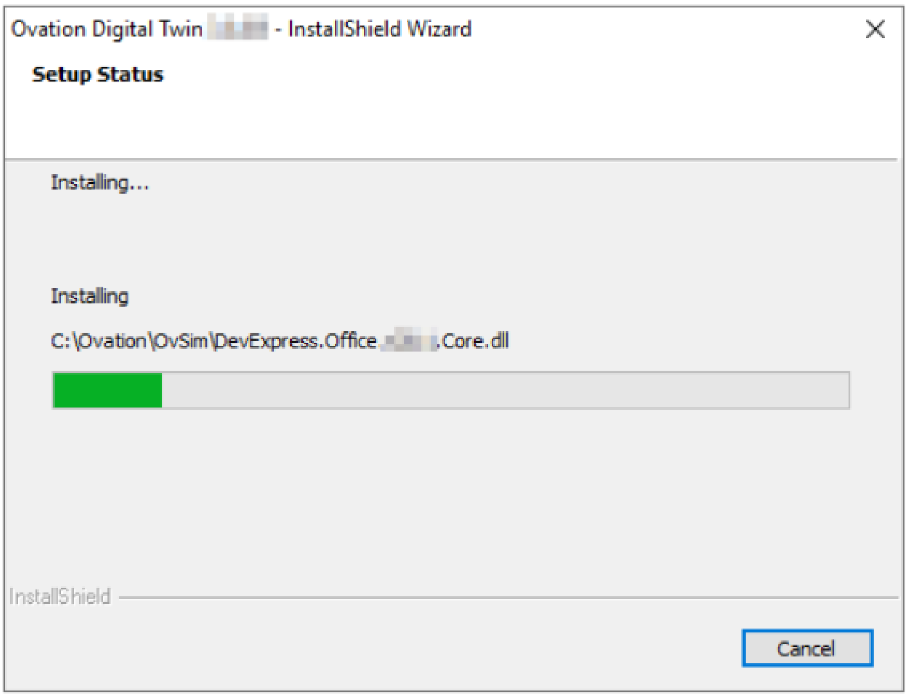
10. The Setup Status page opens, showing the installation progress. The copying progress may take several minutes.

⚠ CAUTION

Several status and processing windows appear during the installation. Do **NOT** dismiss any of these windows since they are normal and necessary for the Ovation Digital Twin installation.

Do **NOT** click the **Cancel** button or the **X** (Close) button on the Setup Status page. Clicking **Cancel** or closing the page will result in improper Ovation Digital Twin installation.

Figure 15. Setup Status page

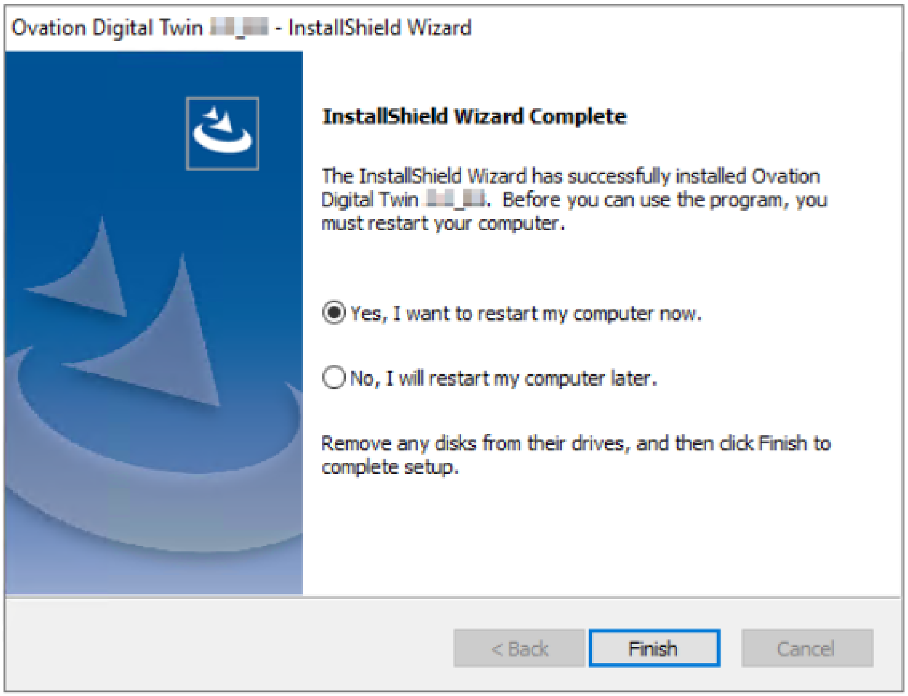


Note

If you must cancel the installation, click **Cancel.** This displays an Exit Setup confirmation box. Click **Yes**. The installation stops and exits.

11. When the installation process is complete, the InstallShield Wizard Complete page appears. For the changes to take effect, you must reboot and login. Select **Yes, I want to restart my computer now** (default selection), then click **Finish**.

Figure 16. InstallShield Wizard Complete page



12. After you install the software on all drops, you must perform the following sequence of actions on all simulation virtual controllers: download, clear, recompile, load. For more information, refer to the *Ovation Developer Studio User Guide*.
13. Delete simalgs.cfg from each DCS drop folder (drop where no simulation models are located) (page 79).

3.2 To install additional Ovation Digital Twin algorithms

Before you attempt to install using the following procedure, make sure that you have obtained a new Algorithms Packages License key code. The new key code contains licenses for your previously obtained algorithms and for your new algorithms. See *To register or renew the Ovation Digital Twin license using the Ovation Digital Twin Tray App or the Ovation Digital Twin LogViewer* (page 16).

Note

Be sure you are installing additional algorithm suites using the same version of the Ovation Digital Twin software as the existing installation.

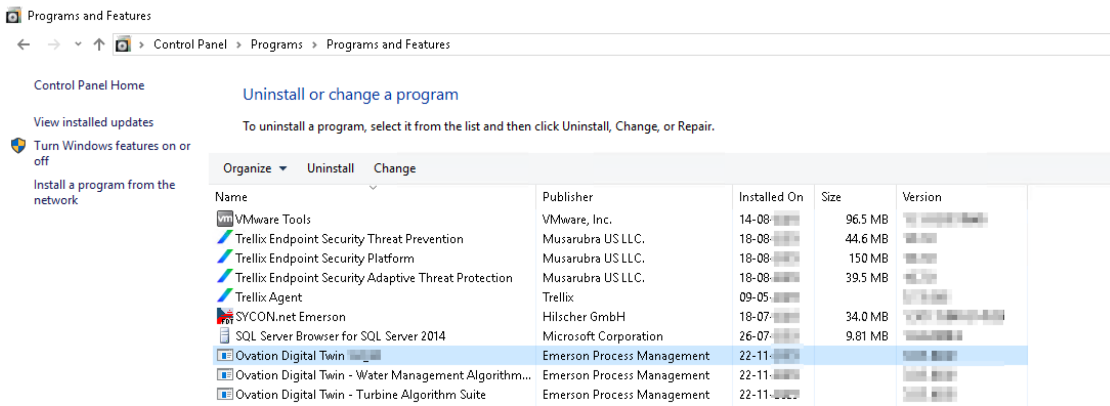
Perform the following steps to install additional Digital Twin algorithms:

1. Insert the Ovation Digital Twin DVD into the DVD drive. If the program does not start automatically, open a Windows Explorer window, access the drive, and double-click **setup.exe**.

   OR

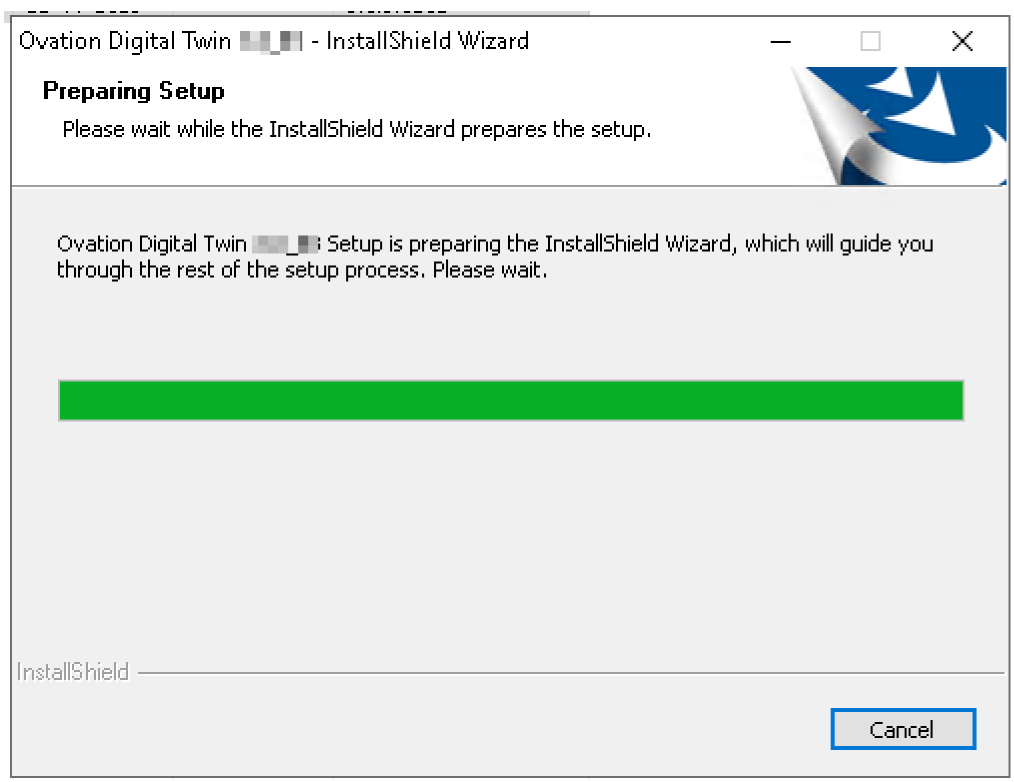
   If you do not have a DVD, navigate to **Start > Control Panel > Programs > Programs and Features.** The Uninstall or change a program screen appears. Select the **Ovation Digital Twin - Base Software** option in the list, then select **Change** from the menu bar.

   Figure 17. Uninstall or change a program window

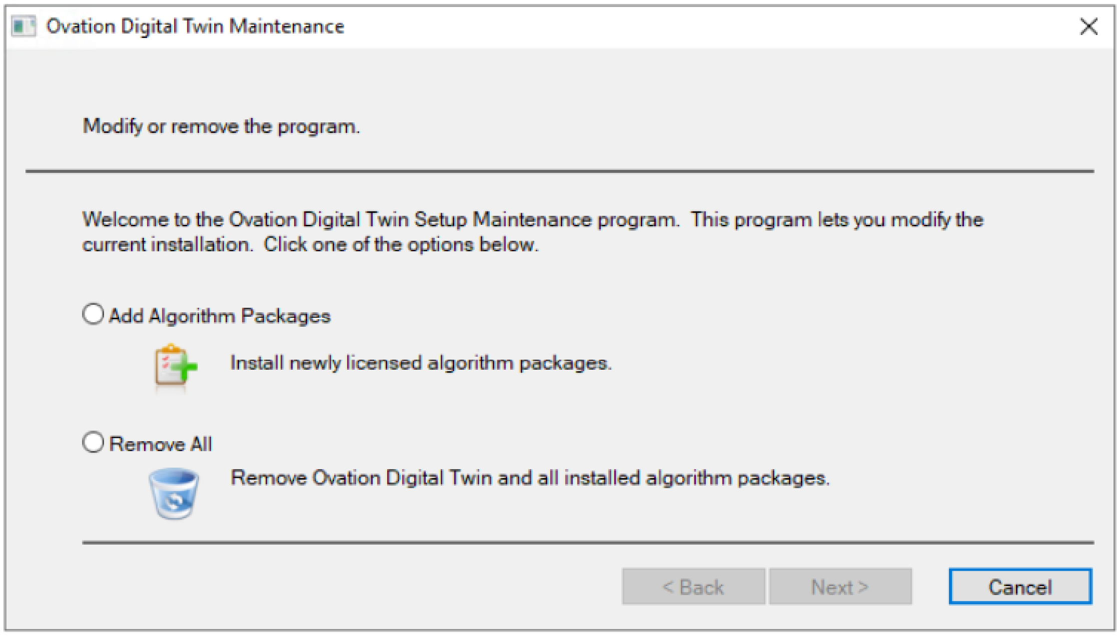
2. The Preparing Setup window appears. No user action is required.

   Figure 18. Preparing Setup window

3. The Modify or remove the program window appears. Select the **Add Algorithm Packages** option, and then click **Next**.

Figure 19. Modify or remove the program window

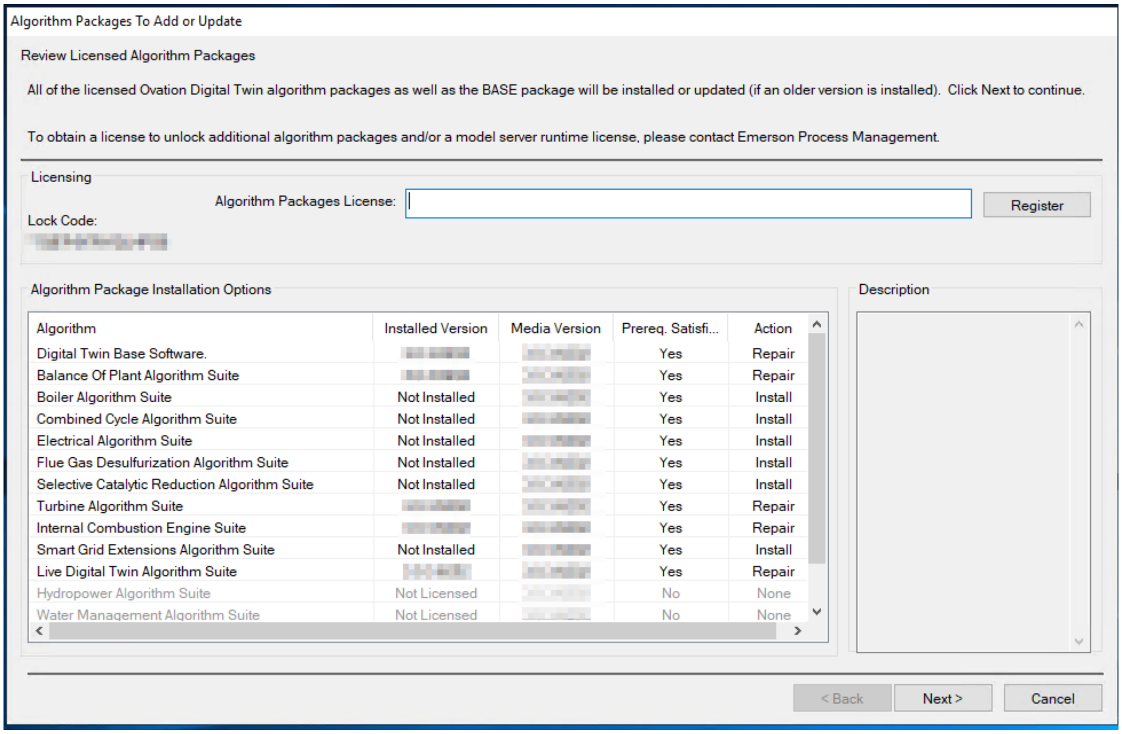


Note

Selecting the **Remove All** option on the Modify or remove the program window uninstalls all Ovation Digital Twin software and algorithms.

4. The Algorithm Packages to Add or Update window appears. Currently installed algorithms (if any) are visible, and all others are grayed out. For example, in the following figure, although the base software is installed, no other algorithm suites are currently installed.
   Enter the new **license key number** in the Algorithm Packages License field, then click **Register**.

   Figure 20. Algorithm Packages to Add or Update window

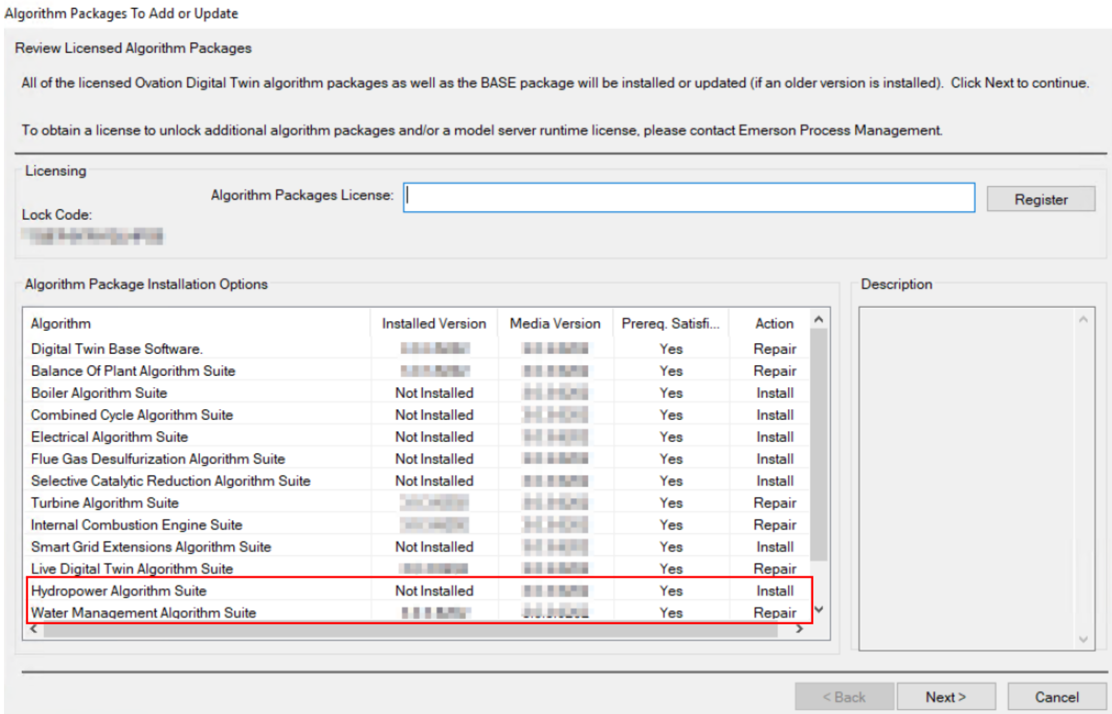
   

   The window updates. The updated list includes the algorithms that are now licensed but not installed, as well as the currently installed algorithms. All other algorithm packages are grayed out. Click **Next**.

   Note

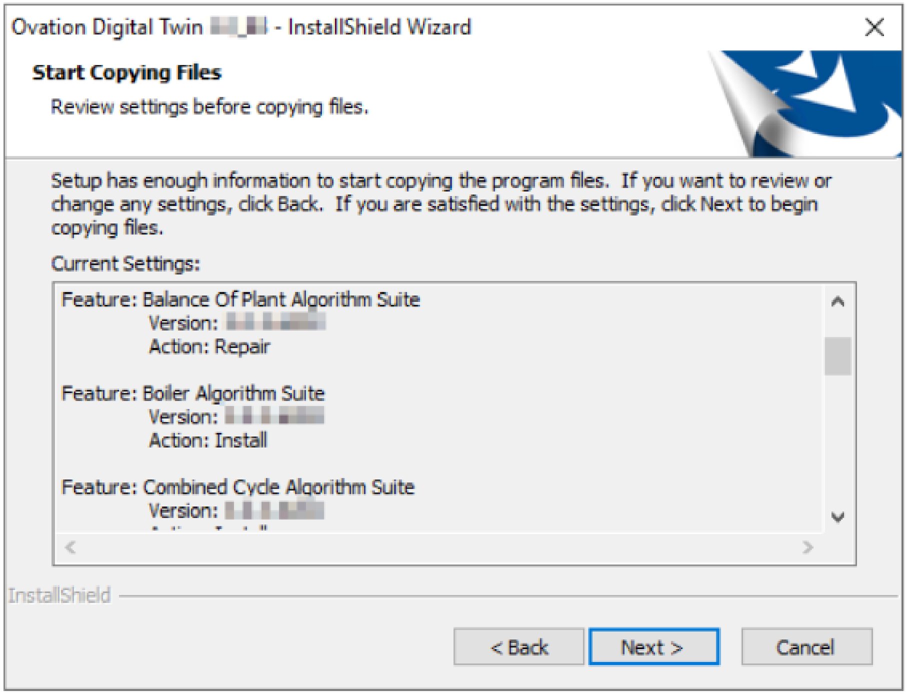
   If you want to install new algorithm suites, you must enter a **license key number** in the License Key entry field. Once you enter the key, the window updates, saying that the algorithm suites are licensed but not installed. For information about obtaining a license key, contact your Emerson service representative.

Figure 21. Algorithm Packages to Add or Update window - algorithm packages registered



5. The Start Copying Files page appears. Review the information, then click **Next**.

Figure 22. Start Copying Files page



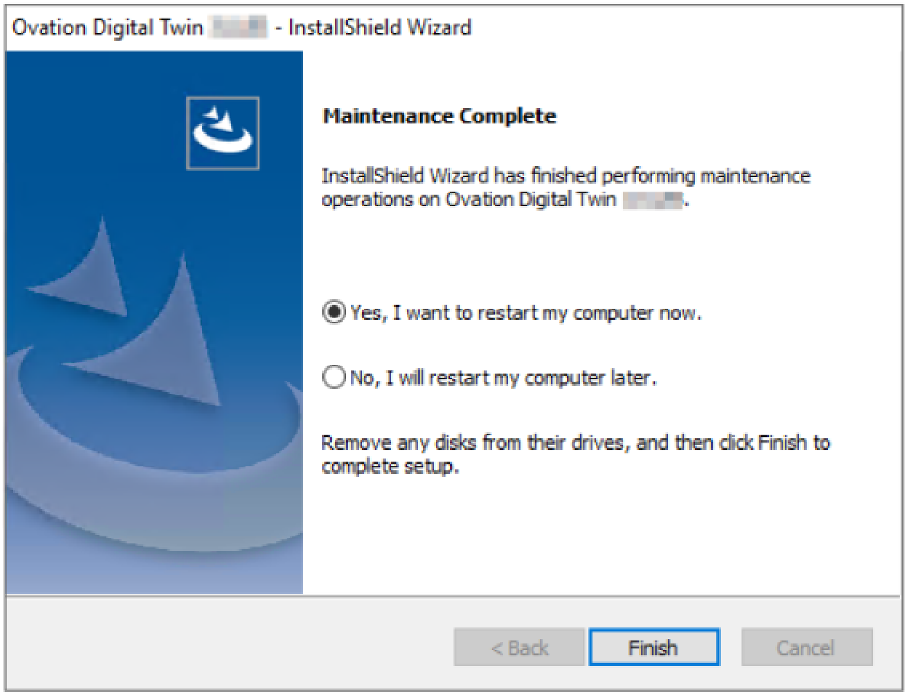
CAUTION

Several status and processing windows appear during the installation. Do **NOT** dismiss any of these windows, as they are normal and necessary for the Ovation Digital Twin Algorithm installation.

Do **NOT** click the **Cancel** button or the **X** (Close) button on the Setup Status window. Clicking **Cancel** or closing the window will result in improper installation of the Ovation Digital Twin algorithms.

6. When the installation is complete, the Maintenance Complete page appears. For the changes to take effect, you must reboot and login. Keep the default selection: **Yes, I want to restart my computer now**, then click **Finish**.

Figure 23. Maintenance Complete page



7. To verify that the algorithm suites were properly installed, navigate to **Start > Control Panel > Programs > Programs and Features**. The newly installed algorithm suites should appear in the window.

3.3 To uninstall the base software and all Ovation Digital Twin algorithm suites

Note

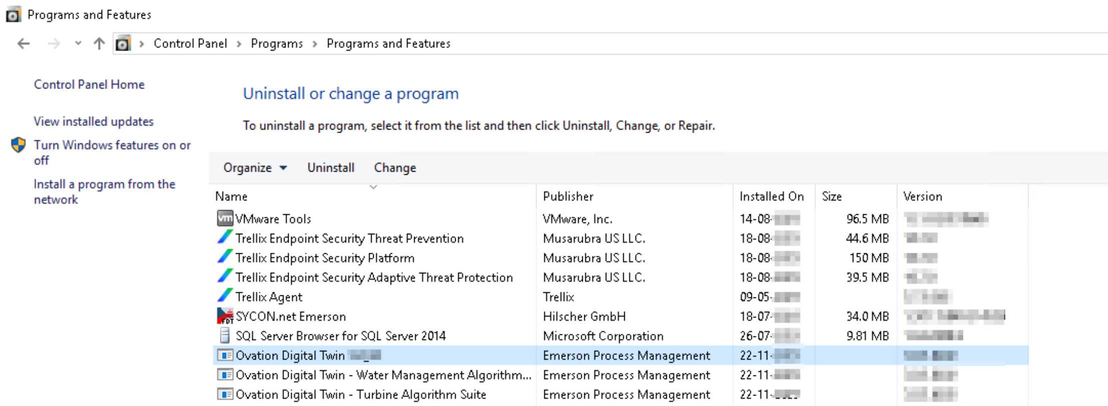
If you uninstall the Ovation Digital Twin base software, you will automatically uninstall all of the installed Ovation Digital Twin algorithm suites.

Perform the following steps to uninstall the Ovation Digital Twin base software and all algorithms:

1. Navigate to **Start > Control Panel**.
2. Click to open **Programs**, then click to open **Programs and Features.**

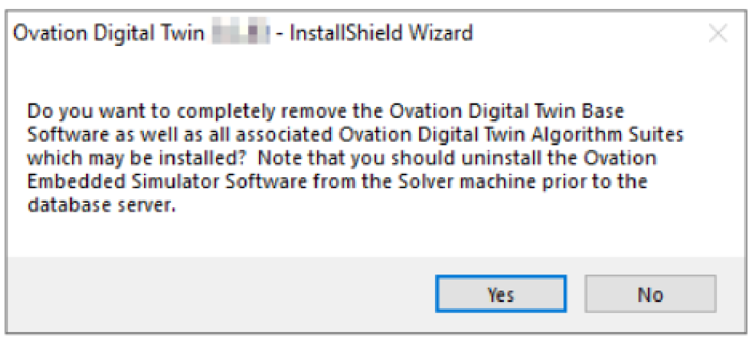
3. The Uninstall or change a program window appears. Select the **Ovation Digital Twin - Base Software** option in the list, then select **Uninstall** from the menu bar.

Figure 24. Uninstall or change a program window



4. The Preparing Setup window appears briefly, and then a confirmation prompt asks you to confirm that you want to remove the base software and all associated algorithm suites. Click **Yes** to continue.

Figure 25. Uninstall confirmation prompt



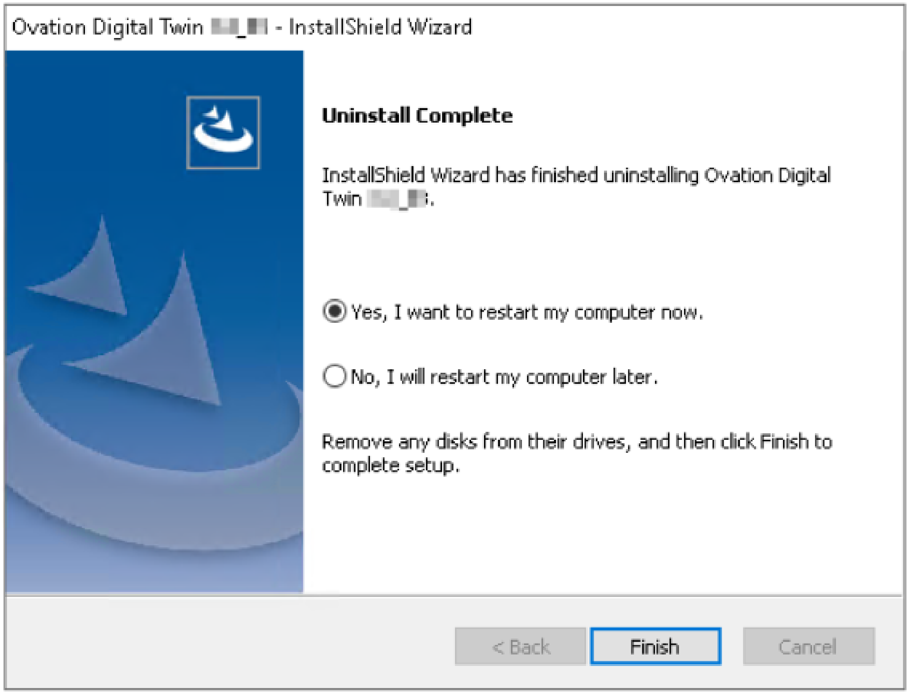
CAUTION

Several status and processing windows appear during the uninstall process. Do **NOT** dismiss any of these windows, as they are normal and necessary for the Ovation Digital Twin uninstallation.

Do **NOT** click the **Cancel** button or the **X** (Close) button on the Setup Status window. Clicking **Cancel** or closing the window results in an improper Ovation Digital Twin uninstallation.

5. When the uninstall is complete, the Uninstall Complete page appears. To reboot, select **Yes, I want to restart my computer now** (default selection), then click **Finish**. The system reboots.

Figure 26. Uninstall Complete page



6. To verify that the base software and algorithm suites were removed, navigate to **Start > Control Panel > Programs > Programs and Features**. The software and algorithm suites no longer appear in the list of installed programs.

4 Using Ovation Digital Twin

Topics covered in this section:

4.1 Overview of using Ovation Digital Twin algorithms in your control scheme

Note

This manual assumes that you have a working knowledge of the Ovation system, including the Ovation Control Builder. For detailed information on Control Builder and other Ovation applications, refer to the *Ovation Control Builder User Guide*, the *Ovation Operator Station User Guide*, or contact your Emerson representative.

To add and use Ovation Digital Twin algorithms in your control scheme, you must work with the Ovation Control Builder.

The following is an overview of typical steps in the process.

1. Access the Ovation Control Builder.
2. Create or open a control sheet and add an Ovation Digital Twin algorithm.
3. Define any required and/or optional parameters using the Control Builder Property Editor.
4. When the sheet is complete, load the Virtual Controller. The sheets are then automatically downloaded by the system to the Operator Stations.
5. Make sure that Ovation Digital Twin is running. Use Ovation Digital Twin Tray App (page 42) or Instructor Station (refer to the *Ovation Digital Twin Instructor Station User Guide*) to start Ovation Digital Twin if it is stopped.
6. Access the Signal Diagram application to view the sheet containing the Ovation Digital Twin algorithm.
7. If the Ovation Digital Twin algorithm is tunable, you may tune it by using the signal diagram Properties Summary window.
8. Once tuning is complete, reconcile the tuning changes between the Controller and the database. Then reconcile the tuning changes between the database and the Control Builder.

5 Ovation Digital Twin Included Applications

Topics covered in this section:

5.1 What are Ovation Digital Twin Included Applications?

When you install Ovation Digital Twin, you also install a collection of applications that work with the Ovation Digital Twin system. Some of them are included with Ovation Digital Twin and some which must be licensed separately.

For details on each of these applications, see the appropriate section of this manual.

Applications which are included with Ovation Digital Twin:

- **Ovation Digital Twin Tray App (page 42)** - Provides basic control of the simulation, version and simulation status display, and access to the Ovation Digital Twin Config, Algorithm Status, and Log Viewer tools.
- **Ovation Digital Twin Config (page 44)** - Allows you to connect the Ovation Digital Twin Tray App to a specific Ovation Digital Twin instance. Once it is connected to a particular simulation, changing its configuration, units, solver parameters, ambient conditions and fuel data is available from this window.
- **Ovation Digital Twin Algorithm Status (page 48)** - Displays information about all algorithms available in the system, as well as navigation to selected algorithm views in other Ovation and Ovation Digital Twin tools.
- **Ovation Digital Twin Log Viewer (page 52)** - Provides access to Ovation Digital Twin logs for troubleshooting purposes.
- **Ovation Digital Twin Network Viewer (page 53)** - Displays the flow, electrical, and mechanical networks which are included in the simulation. Provides a more advanced look into runtime conditions in the simulation and the underlying model structure.

- **Ovation Digital Twin Advanced Edit (page 57)** - When accessed from a selected algorithm, it provides customization, recalculation, import/export and notes features specific to that algorithm.
- **Ovation Digital Twin Advanced Tuning (page 65)** - Displays algorithm point name and system ID (SID) information and allows for easy navigation to other Ovation and Ovation Digital Twin tools. It provides a specialized, algorithm-specific view of the algorithm states, beyond the simple output parameters of Signal Diagram. For some algorithms, the ATI also provides a means to adjust select tuning parameters and/or state variables.
- **Ovation Digital Twin Media Tables (page 69)** - Allows you to use Ovation Digital Twin built-in property tables for all media supported by Ovation Digital Twin.
- **Ovation Digital Twin Control Builder Unit Converter (page 73)** - Provides a specialized interface for converting engineering units used in control algorithms within the Ovation Digital Twin environment.

Applications which require a separate license:

- Ovation Digital Twin Spreadsheet Tool
- Ovation Digital Twin Instructor Station
- Ovation Digital Twin Training Kiosk
- Ovation Digital Twin third-party interfaces

This section discusses applications which are included with Ovation Digital Twin.

Note

For more information on Ovation Digital Twin Spreadsheet Tool, refer to the *Ovation Digital Twin Tools User Guide* or consult your Emerson representative.

For more information on Ovation Digital Twin Instructor Station and Ovation Digital Twin Training Kiosk, refer to the *Ovation Digital Twin Instructor Station User Guide* or consult your Emerson representative.

5.2 Ovation Digital Twin Tray App

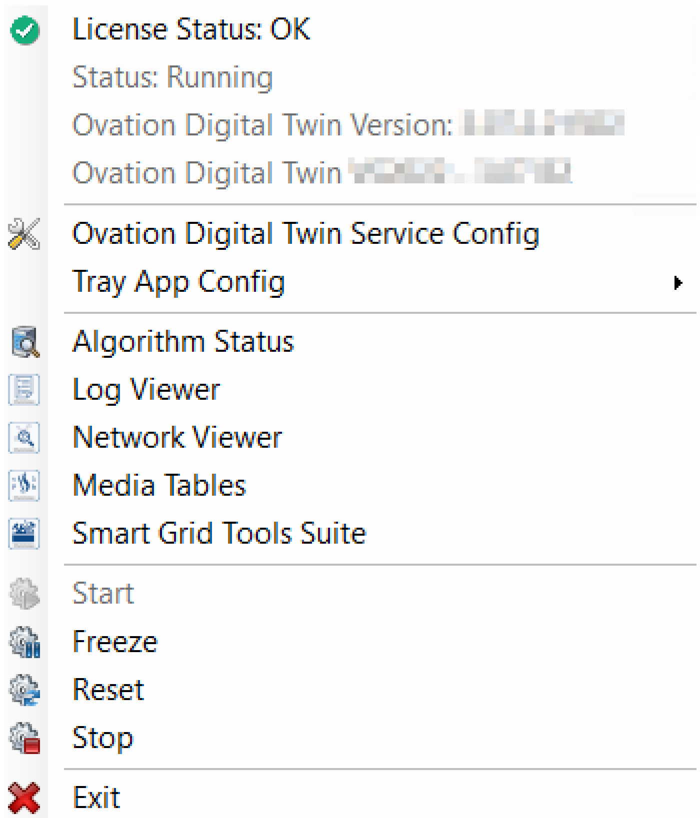
The Ovation Digital Twin Tray Application (Ovation Digital Twin Tray App) gives you an overview of the connected simulation, plus simple tools for controlling it.

Additionally, it provides easy access to other built-in Ovation Digital Twin tools.

Figure 27. Tray App desktop icon

Navigation in the Ovation Digital Twin Tray App is accessed through a control menu, which you can access by right-clicking the Ovation Digital Twin Tray App icon in the Windows taskbar in the bottom right of your desktop or navigating to C:\Ovation\OvSim\OvSim.TrayApp.exe.

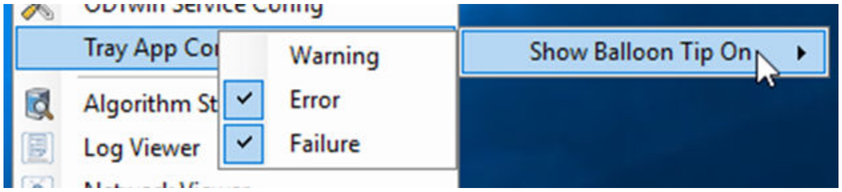
Figure 28. Ovation Digital Twin Tray App control menu



The Ovation Digital Twin Tray App control menu provides status information and control functions:

- **License Status** - Opens the License Information window (page 17).
- **Status** - Displays the current status of the simulation (such as Running, Frozen, Stopped, or Disconnected).
- **Ovation Digital Twin version** - Displays the auto-generated version number of the Ovation Digital Twin software application.
- **Ovation Digital Twin - X.XX XX** - Displays the version number of the current simulation.
- **Ovation Digital Twin Service Config** - Opens the Global Parameters Configuraton window (page 44).
- **Tray App Config** - Opens the Show Balloon Tip On menu. To enable a pop-up message to display for Warning, Error, or Failure severity events, click to select the appropriate option(s).

Figure 29. Show Balloon Tip on function



- **Algorithm Status** - Opens the Ovation Digital Twin Algorithm Status tool (page 48).
- **Log Viewer** - Opens the Ovation Digital Twin LogViewer tool (page 52).
- **Network Viewer** - Opens the Ovation Digital Twin Network Viewer tool (page 53).
- **Media Tables** - Opens the Ovation Digital Twin Media Tables tool (page 69).
- **Smart Grid Tools Suite** - Opens the SGE Tools Suite Application (STSA) and SGE Hosting Capacity Study (HCS). For more information, refer to the *Ovation Digital Twin Smart Grid Extensions Suite User Guide*.
- **Start** - Starts the simulation.
- **Freeze** - Pauses the simulation.
- **Reset** - Clears progress and resets the simulation to the beginning state.
- **Stop** - Stops the simulation.
- **Exit** - Closes the Ovation Digital Twin Tray App.

5.2.1 Ovation Digital Twin Config

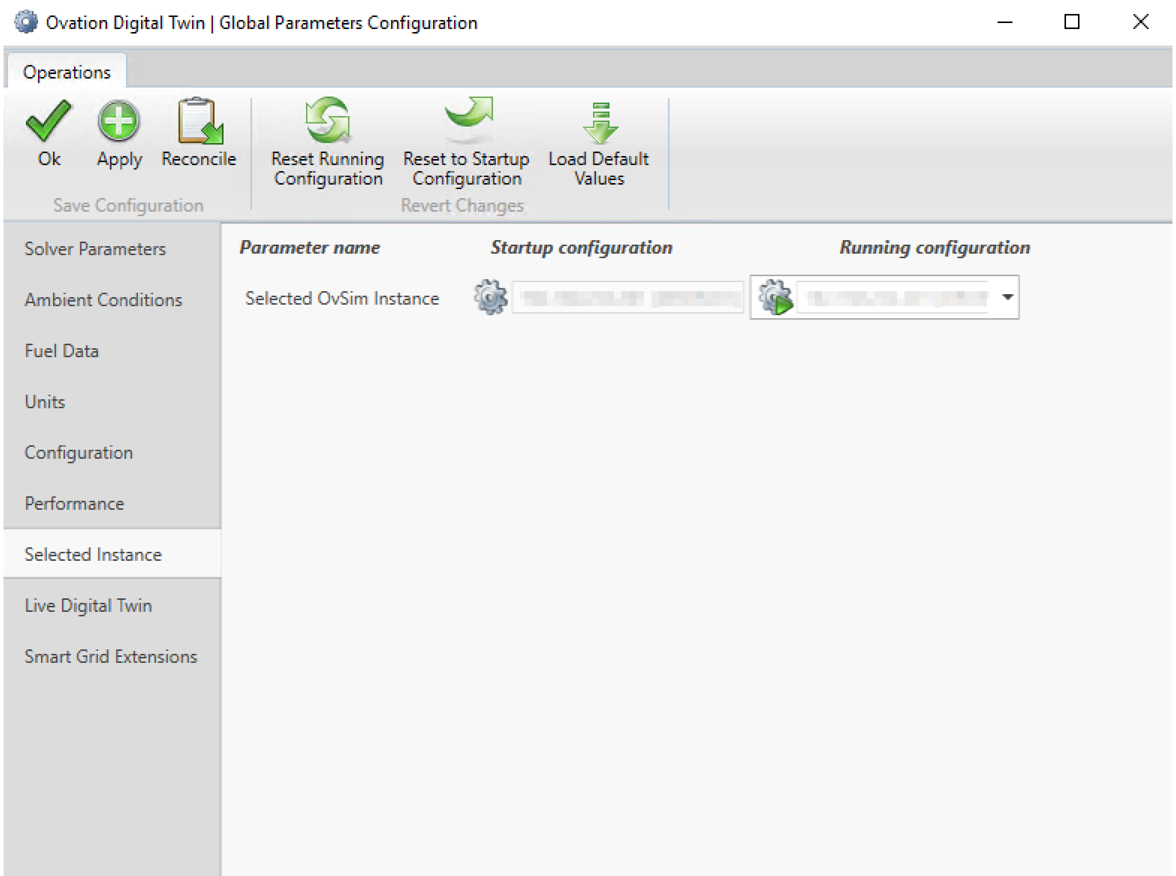
Ovation Digital Twin Config allows you to adjust the configuration of a specific Ovation Digital Twin instance. Once the Ovation Digital Twin Tray App connects to a particular simulation, you can use the Ovation Digital Twin Tray App menu to access the Ovation Digital Twin Config interface, the Global Parameters Configuration window.

In the Global Parameters Configuration window, you can see and change the different simulation's configuration options.

To access the Global Parameters Configuration window, open the Ovation Digital Twin Tray App and click the **Config** menu option.

The Global Parameters Configuration window opens to the Selected Instance tab.

Figure 30. Ovation Digital Twin Tray App Global Parameters Configuration window - Selected Instance tab selected



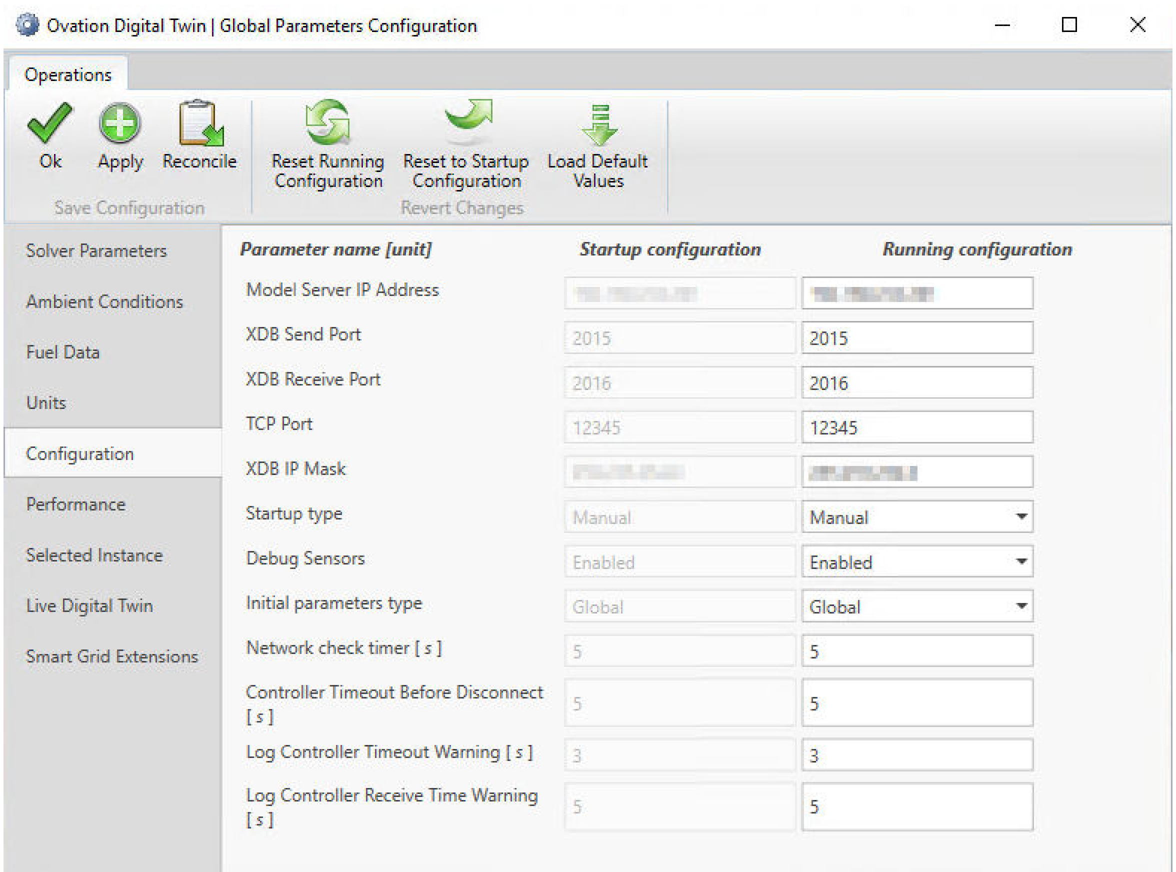
Tabs running down the left side of the window provide access to the various areas of the simulation:

- **Solver Parameters** - Configures the Ovation Digital Twin flow, electrical and mechanical networks solvers parameters.
- **Ambient Conditions** - Allows you to configure globally used ambient condition parameters.
- **Fuel Data** - Allows you to configure globally used fuel parameters.
- **Units** - Allows you to choose between available unit sets.
- **Configuration** - Allows you to configure Ovation Digital Twin connectivity.
- **Performance** - Allows you to **configure performance related parameters**. Additionally, it shows basic system performance metrics. It also allows you to capture **performance diagnostic data (with detailed breakdown of individual tasks and algorithms execution time and threading)**. It also allows comparison of previously collected performance diagnostic data.
- **Selected Instance** - Allows you to choose which simulation instance the Ovation Digital Twin Tray App is connected to and controls. Typically, only one simulation instance is present in the system.
- **Live Digital Twin** - Enables configuration specific to the Live Digital Twin simulation.

- **Smart Grid Extensions** - Enables configuration specific to the Smart Grid Extensions suite of algorithms.

Selecting different tabs displays information specific to that parameter.

Figure 31. Ovation Digital Twin Tray App Global Parameters Configuration - Configuration tab selected



The main area of the window displays basic information about the parameter for the tab you selected:

- **Parameter name** - Name of the parameter.
- **Startup configuration** - Saved value of the parameter.
- **Running configuration** - Current value of the parameter. You can change this value without reconciling it, so that you can later return to the original startup configuration value.

A ribbon tab located along the top of the window provides access to the following functions:

- **OK** - Applies the running configuration to the simulation and exits.
- **Apply** - Applies the modified running configuration to the simulation.
- **Reconcile** - Saves the running configuration as startup configuration.
- **Reset running configuration** - Removes unapplied modifications to the running configuration.
- **Reset to startup configuration** - Overrides the running configuration with startup configuration.

- **Load default values** - Overrides the running configuration with default configuration.

5.3 Ovation Digital Twin Algorithm Status

The Ovation Digital Twin Algorithm Status app displays the status and other information about all algorithms available in the system.

From this app, you can:

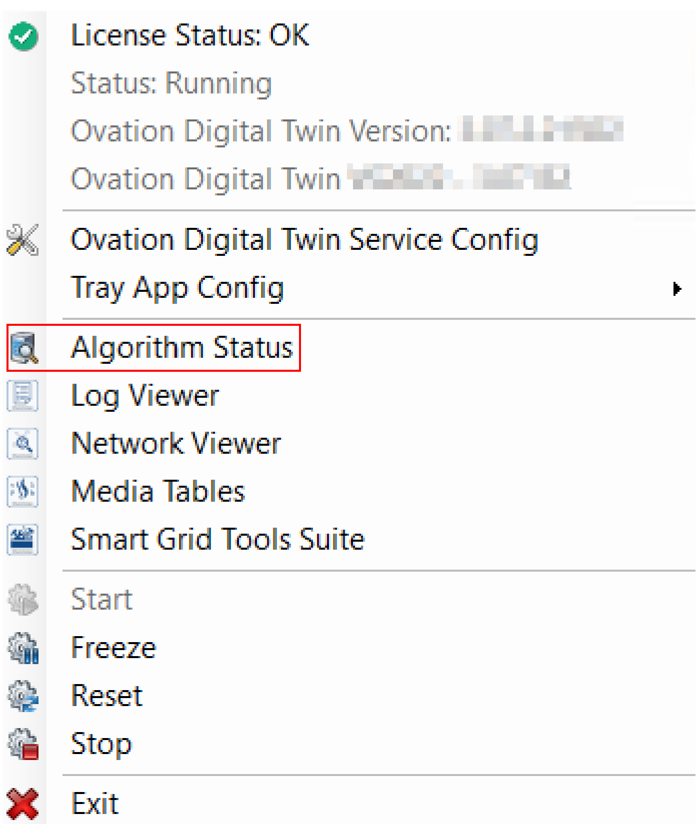
- View the status information on all algorithms.
- Locate problematic algorithms for troubleshooting.
- View basic or detailed information about a selected algorithm.
- Search and filter algorithms.
- View the selected algorithm in another Ovation (Control Builder, Signal Diagram, Point Info) or Ovation Digital Twin (Network Viewer, Log Viewer) application.

To access the Ovation Digital Twin Algorithm Status app, right-click the **Ovation Digital Twin Tray App** in the Windows taskbar in the bottom right of your desktop, then click on the Algorithm Status menu option.

Note

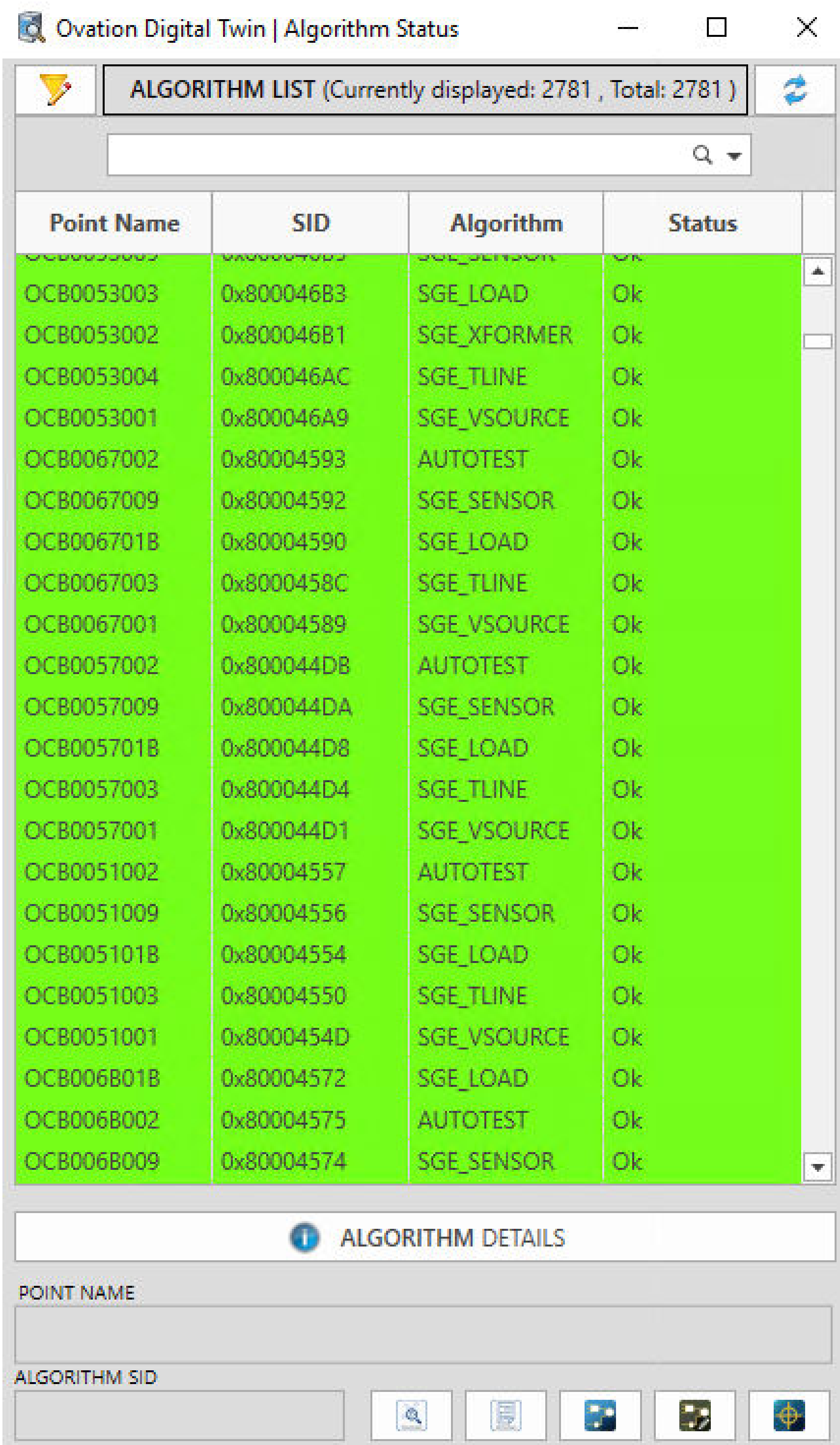
You can also launch the Algorithm Status app from the Log Viewer, Network Viewer, and Advanced Tuning tools.

Figure 32. Ovation Digital Twin Tray App menu options- Algorithm Status



The Ovation Digital Twin Algorithm Status window opens.

Figure 33. Algorithm Status



The Algorithm Status lists all algorithms by:

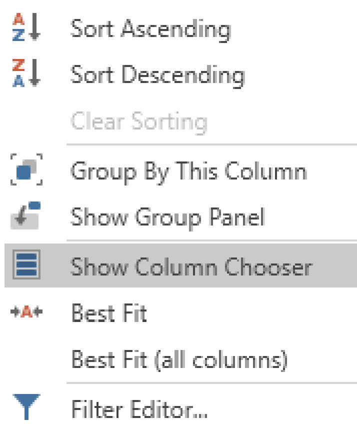
- **Point Name** - Name of the Ovation point attached to the algorithm.

- **SID** - System identification for the point.
- **Algorithm** - Algorithm model name.
- **Status** - Current status of the algorithm (such as Ok, Warning, and so forth).

Note

To customize the Algorithm List Filter, right-click on a column header and use the context menu to change filter options, hide or display columns, sort, group, and so forth.

Figure 34. Algorithm List Filter context menu



To find a specific algorithm, either scroll to it or type the algorithm name in the search field.

To examine a particular algorithm more closely, highlight the algorithm in the list. The window below the list displays the algorithm Point Name and Algorithm SID.

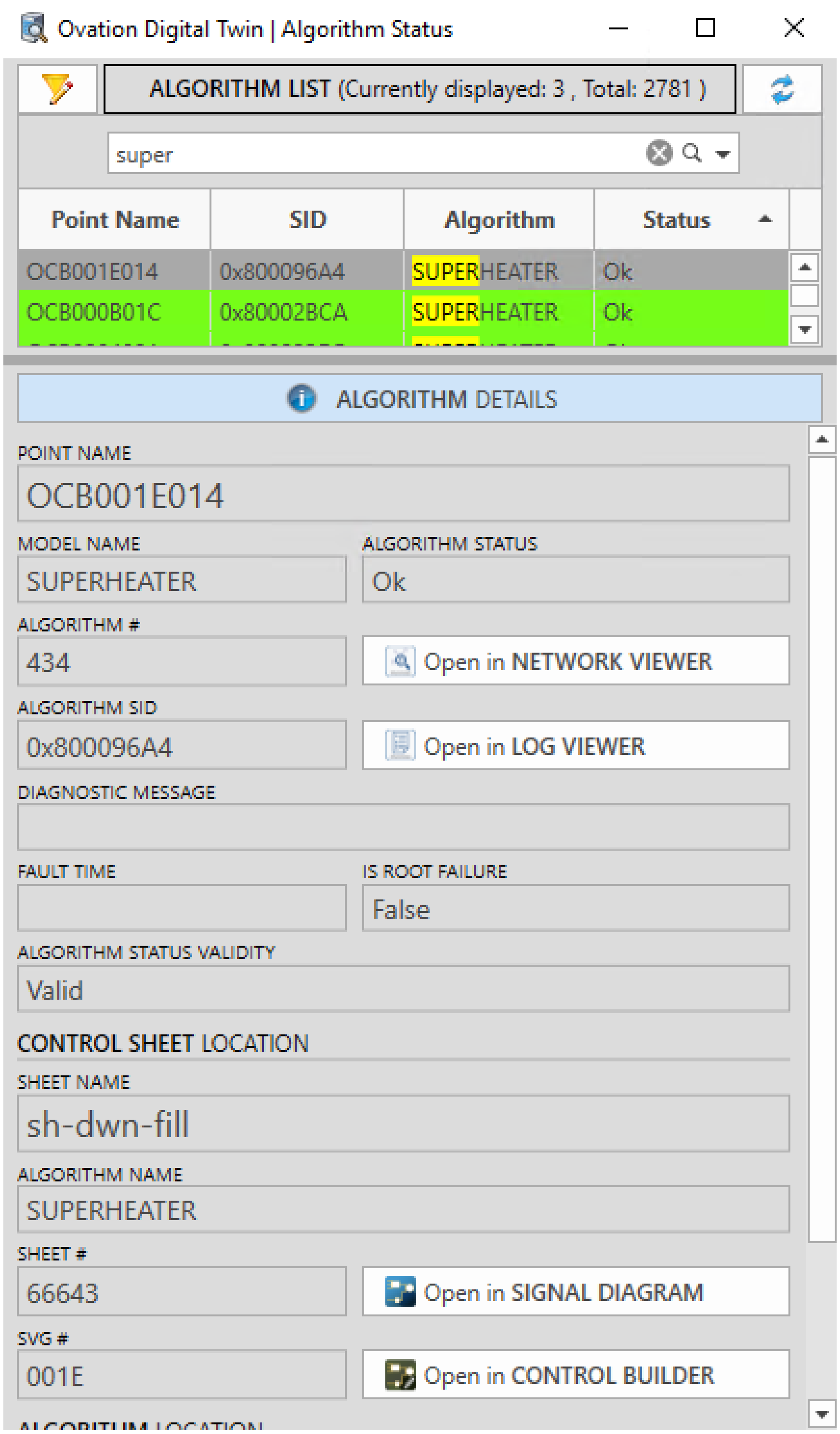
Five icons to the right of the Algorithm SID allow you to open the algorithm in other Ovation or Ovation Digital Twin applications. Hover and click the applicable icon button to open the Ovation Digital Twin Network Viewer (page 53), Ovation Digital Twin Log Viewer (page 52), Ovation Signal Diagram, Ovation Control Builder, and/or Ovation Point Information applications.

Note

Network Viewer and Log Viewer are included in the Ovation Digital Twin installation and are discussed in this manual. For information on using Signal Diagram, Control Builder, or Point Info, refer to the *Ovation Control Builder User Guide*, *Ovation Operator Station User Guide*, or contact your Emerson representative.

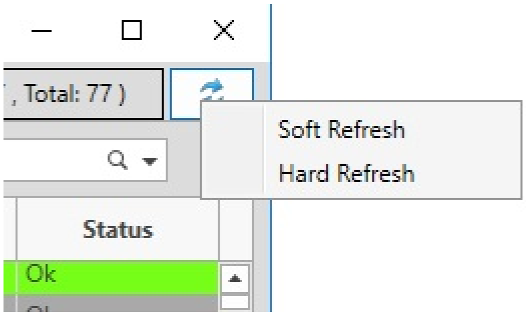
To see more details on a particular algorithm, double-click on the algorithm in the list to open the Algorithm Details window.

Figure 35. Algorithm Details window



To refresh the results, click the blue **Refresh** button in the upper-right corner of the window. Select **Soft Refresh** to reset all filters/searches/groups to default while maintaining cached point information. Select **Hard Refresh** to clear the cached point information and reset the Algorithm Status List to perform another algorithm scan, especially if Ovation point data has changed.

Figure 36. Refresh button



5.4 Ovation Digital Twin LogViewer

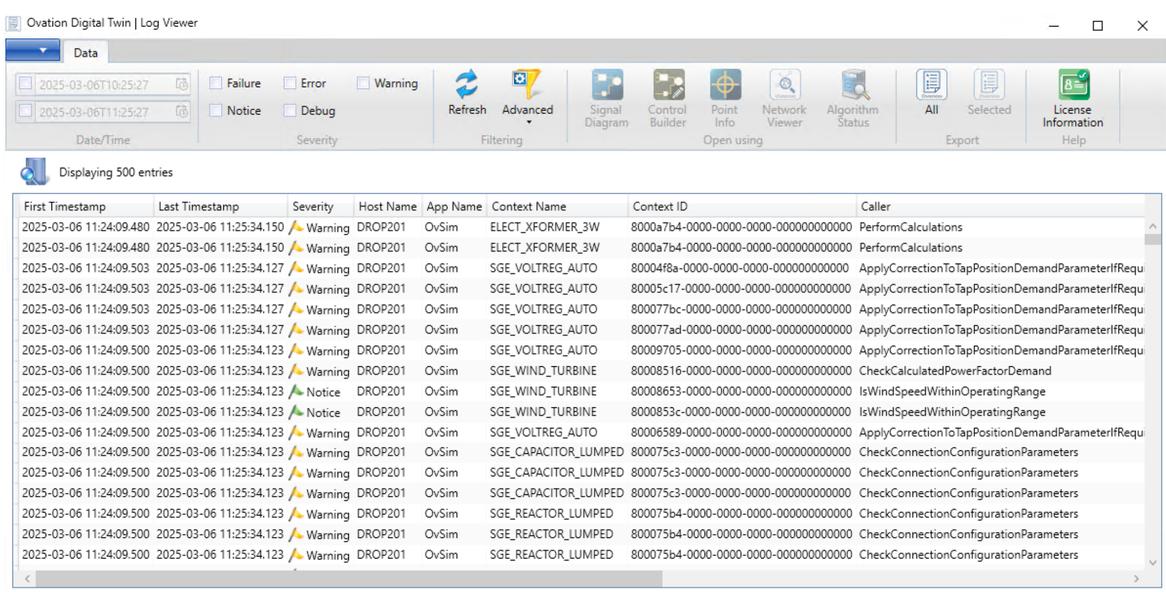
The Ovation Digital Twin LogViewer allows you to access Ovation Digital Twin logs for troubleshooting. You can filter logs, export them, and open applications specific to the algorithm associated with a particular log entry.

To open the Ovation Digital Twin LogViewer, either double-click the **Ovation Digital Twin LogViewer** icon on your desktop or navigate to C:\Ovation\OvSim\OvSim.LogViewer.exe.

Figure 37. Ovation Digital Twin LogViewer icon

The Ovation Digital Twin LogViewer window opens.

Figure 38. Ovation Digital Twin LogViewer window



Log entries include the following information:

- **First Timestamp** - Time when event happened. (first time if quantity > 1)
- **Last Timestamp** - Time when event happened. (last time if quantity > 1)
- **Severity** - Degree of severity: Failure, Error, Warning, Notice, Debug.
- **Host Name** - Name of machine, e.g. DROP200.
- **App Name** - Application name: OvSim.
- **Context Name** - Source of log event.
- **Context ID** - Source ID. For models, the first 8 hexadecimal digits correspond to the Ovation point SID with remaining digits for internal use.
- **Caller** - Name of specific method causing log event (for developer use only).
- **LineNumber** - Log location (for developer use only).
- **Message** - Description of event.
- **Quantity** - Number of times an event has happened during timestamp period.

A ribbon tab along the top of the Log Viewer provides the following functions:

- **Date/Time** - Use the date fields to filter log results for a range of dates.
- **Severity** - Mark the boxes to filter log results for one or more severity levels.
- **Filtering** - Click the **Refresh** button to re-filter results on adjusted criteria. Click the **Advanced** button to add new criteria to filter.
- **Open using** - Click the icons to open the applications Signal Diagram, Control Builder, Point Info, Ovation Digital Twin Network Viewer, or Ovation Digital Twin Algorithm Status.

  If a selected log corresponds to a particular algorithm, the navigation icons are active and the applications (Signal Diagram, Point Info, and so forth) open on the algorithm responsible for this particular log. If the selected log is of a generic nature, the navigation icons are disabled (grayed out).

  Note

  Ovation Digital Twin Network Viewer and Ovation Digital Twin Algorithm Status are included in the Ovation Digital Twin installation and are discussed in this manual. For information on using Signal Diagram, Control Builder, or Point Info, refer to the *Ovation Control Builder User Guide*, the *Ovation Operator Station User Guide*, or contact your Emerson representative.

- **Export** - To export all results to a spreadsheet, click the **All** button. To export selected results to a spreadsheet, highlight the results to export, then click the **Selected** button. Logs are saved in C:\Ovation\OvSim\Log.
- **License Information** - Opens the *License Information window* (page 17).

5.5 Ovation Digital Twin Network Viewer

The Ovation Digital Twin Network Viewer provides a more advanced look into the runtime conditions in the simulation as well as the underlying structure of the models. It displays the networks which comprise the simulation – flow, electrical and mechanical parts.

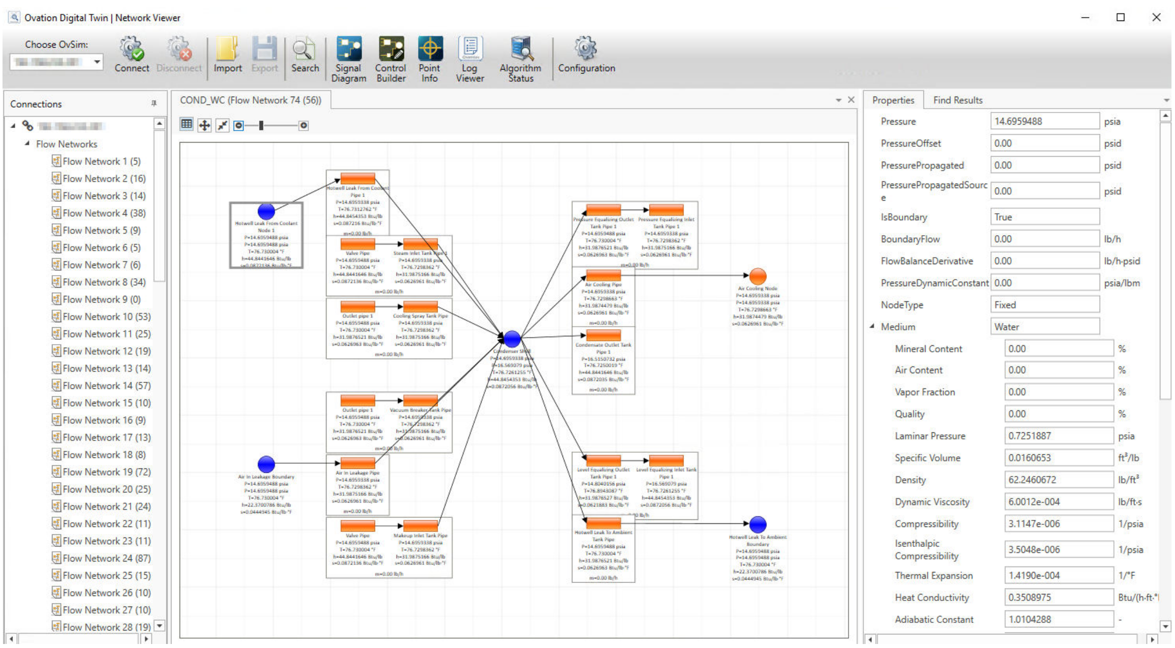
When an element is selected, a runtime properties display appears, providing navigation features, as well as advanced filtering/search and import/export features.

To open the Ovation Digital Twin Network Viewer, either double-click the **Ovation Digital Twin Network Viewer** icon on your desktop or navigate to C:\Ovation\OvSim\OvSim.NetworkViewer.exe.

Figure 39. Ovation Digital Twin Network Viewer icon

The Network Viewer window opens.

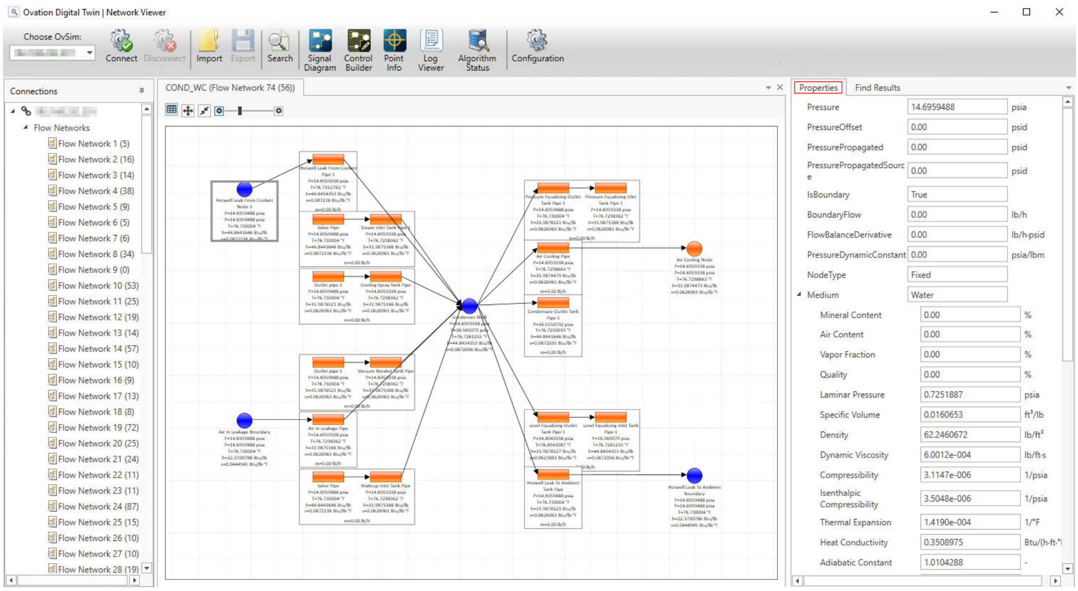
Figure 40. Network Viewer window



The Network Viewer window is divided into three panes:

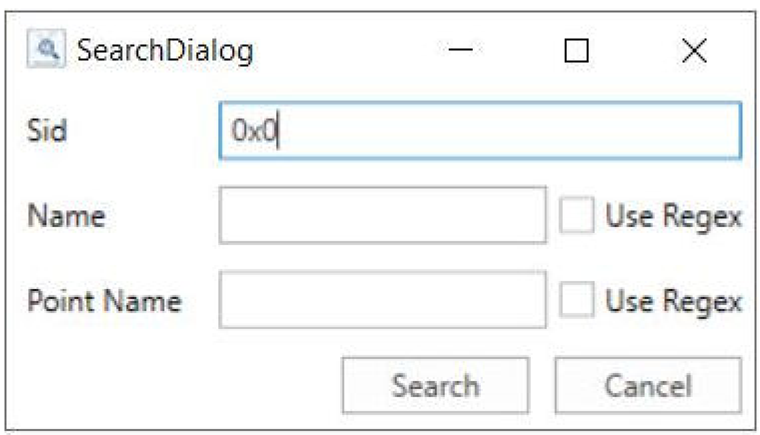
- **Left pane (Directory tree)** - Navigates to the network you want to see.
- **Center pane (Network diagram)** - Displays the network diagram for the network you selected from the directory tree.
- **Right pane (Properties and Find Results)** - Consists of two tabs:
    - **Properties** - Displays the properties of the selected algorithm piece.

Figure 41. Network Viewer window- Properties Tab



◦ **Find Results** - Displays the results of a search performed using the Search button in the ribbon toolbar. The search results are organized into two main sections- Networks and Algorithms. Additionally, if you open the Network Viewer from Algorithm Status, Log Viewer, or Advanced Tuning, the Find Results tab will automatically populate with the information about the algorithm from which you have opened it.

Figure 42. SearchDialog box

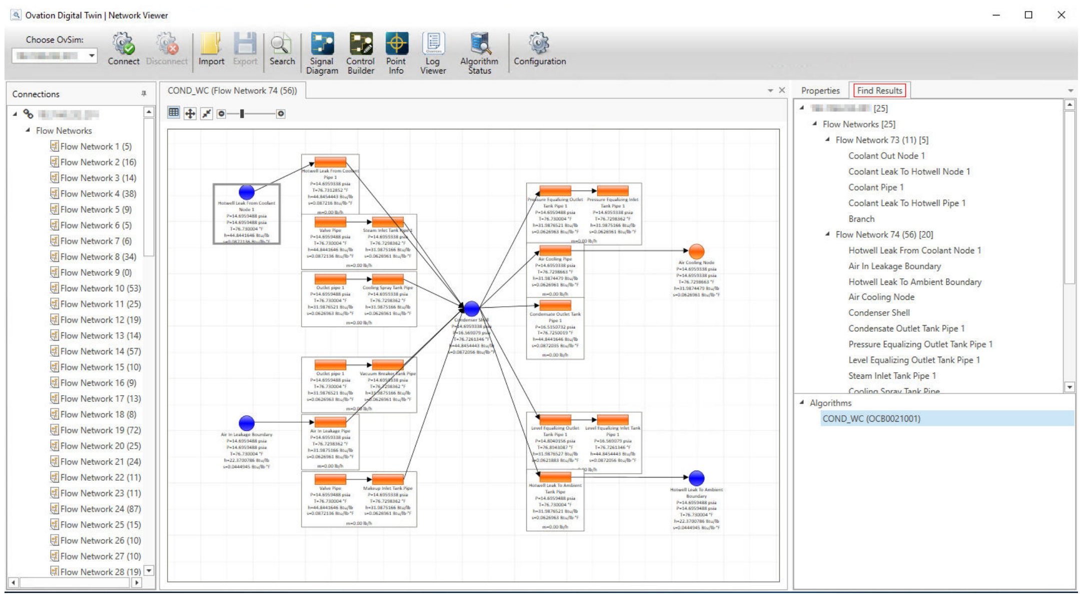


The Find Results tab offers two panes:

– **Networks (Upper pane)** - Shows all the internal pieces of the selected network (limited by the tree depth configuration for larger networks).

– **Algorithms (Lower pane)** - Shows all the internal pieces for the selected algorithm.

Figure 43. Network Viewer window- Find Results Tab



A ribbon tab at the top of the Network Viewer provides the following functions:

- **Choose OvSim** - Chooses the simulation instance to connect to. Usually, only one simulation instance is in a system.
- **Connect/Disconnect** - Connects and disconnects the simulation.
- **Import** - In offline mode, you can import a previously saved network.
- **Export** - Exports the network data to a file for future reference.
- **Search** - Allows searching for an algorithm of interest within all the simulation networks.
- **Applications** - Allows you to navigate to the Ovation Digital Twin Log Viewer, Ovation Digital Twin Algorithm Status, Signal Diagram, Control Builder, and Point Info applications.

Note

Ovation Digital Twin Log Viewer and Ovation Digital Twin Algorithm Status are included in the Ovation Digital Twin installation and are discussed in this manual. For information on using Signal Diagram, Control Builder, or Point Info, refer to the *Ovation Control Builder User Guide*, *Ovation Operator Station User Guide*, or contact your Emerson representative.

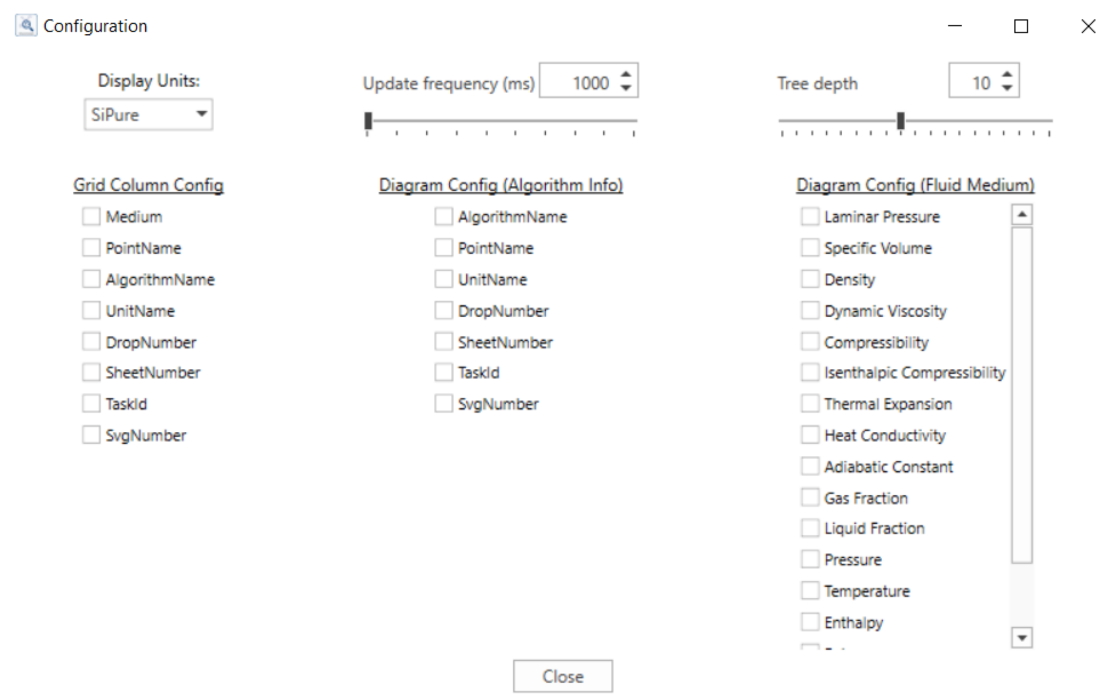
- **Configuration** - Allows you to change the following parameters:
  - **Display Units** - A dropdown where you choose the unit system.
  - **Update Frequency** - A slider and input box to set how frequently (in milliseconds) the system fetches data from the OvSim service and refreshes the network view. A lower number means more frequent updates.
  - **Tree Depth** - A slider to set how many branches appear in the network from the selected network element.

  You can also specify which data should be displayed at nodes, from Algorithm Information to Medium Properties (AlgorithmName, DropNumber, Density, Enthalpy, and so forth).

CAUTION

Faster update frequency, higher tree depth, and more displayed properties will cause higher network traffic on the system and may affect the simulation performance.

Figure 44. Configuration window



5.6 Ovation Digital Twin Advanced Edit

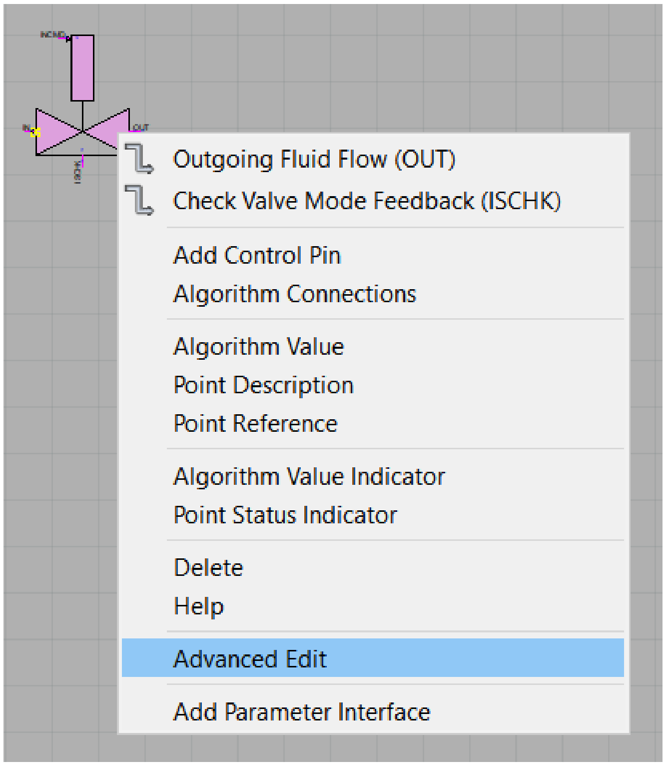
The Ovation Digital Twin Advanced Edit application provides import/export options and a unit recalculation function. It is accessible from Ovation Control Builder. Additionally, select algorithms have a more advanced, custom edit user interface available.

Note

For information on accessing and using Ovation Control Builder, refer to the *Ovation Control Builder User Guide* or contact your Emerson representative.

To access Ovation Digital Twin Advanced Edit, right-click on an Ovation Digital Twin algorithm location in Control Builder, then select **Advanced Edit** from the popup menu.

Figure 45. Advanced Edit menu



The Advanced Edit window opens.

The left pane includes a list of all parameters (with units) of the selected algorithm.

A ribbon tab located along the top of the window provides access to the following functions:

- **Save Configuration**
  - **OK** - Exits the Advanced Edit application and updates all modified algorithm tuning values in active Control Builder sheet. The control sheet must be saved to apply changes.
  - **Reset** - Discards all modified algorithm tuning values and resets all parameters to the values when Advanced Edit application was opened.
  - **Cancel** - Exits the Advanced Edit application without updating any modified algorithm tuning values in the active Control Builder sheet.
- **Import/Export to file**
  - **Import** - Imports an external file containing all algorithm tuning values for the algorithm into the current Advanced Edit instance.
  - **Export** - Creates an external file containing all algorithm tuning values for the current algorithm.
- **Select Units** - Converts algorithm tuning values from the active unit set to the selected unit set. Used when the unit set for tuning parameters differs from the units selected in the Ovation Digital Twin Global Parameters Configuration.
  - **SI Pure** - Converts all algorithm tuning values from current units to pure SI units.

- **SI Engineering** - Converts all algorithm tuning values from current units to Engineering SI units.
- **English** - Converts all algorithm tuning values from current units to English units.

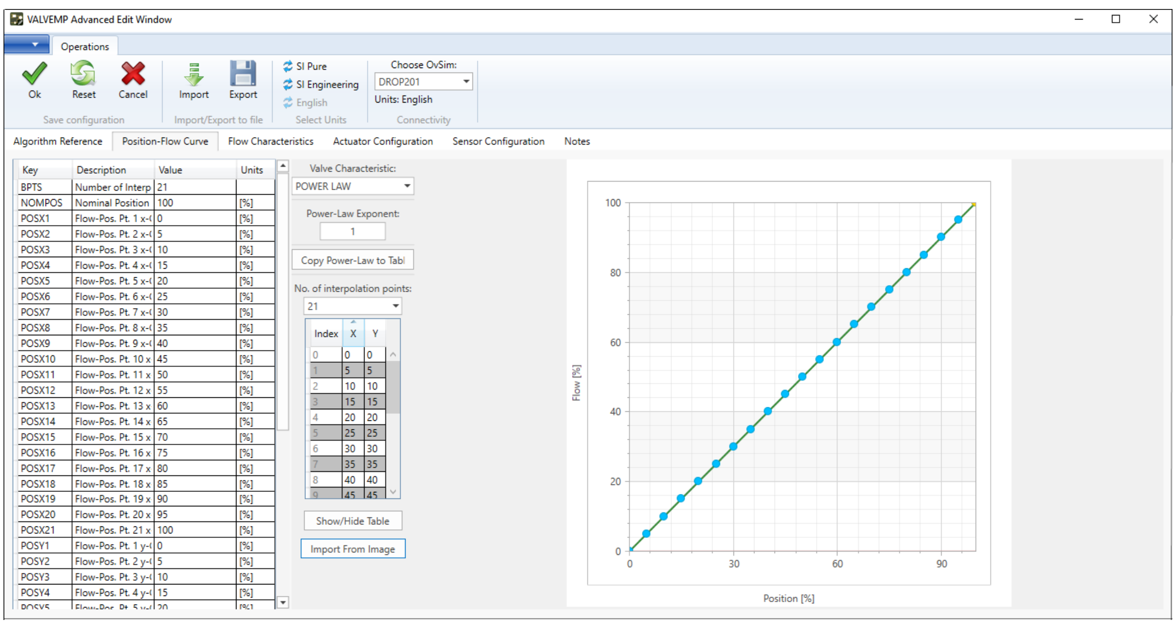
WARNING

This unit conversion is useful when importing control sheets from different simulators that have different unit sets. Before you start running your simulation, you should make sure that all of the algorithms have been converted to (or tuned using) the same unit set - the one selected in the Ovation Digital Twin Global Parameters Configuration.

- **Connectivity** - Selects the local Ovation Digital Twin Model Server instance to use for configuration loading. Relevant Model Server resources include the configured Ovation Digital Twin unit set and the Model Configuration files.
  - **Choose OvSim** - Sets the specified Ovation Digital Twin Model Server as the active instance for configuration resources.

Depending on the algorithm, a set of additional parameterization tabs appear (for example, Position-Flow Curve, Flow Characteristics, Actuator Configuration, Sensor Configuration, etc.).

Figure 46. Advanced Edit window- Position-Flow Curve



5.6.1 Graph Digitizer

For algorithms supporting characteristic curves, the Import From Image button allows you to digitize characteristic curve data and import the data directly for each algorithm.

The following list describes the Graph Digitizer functionalities:

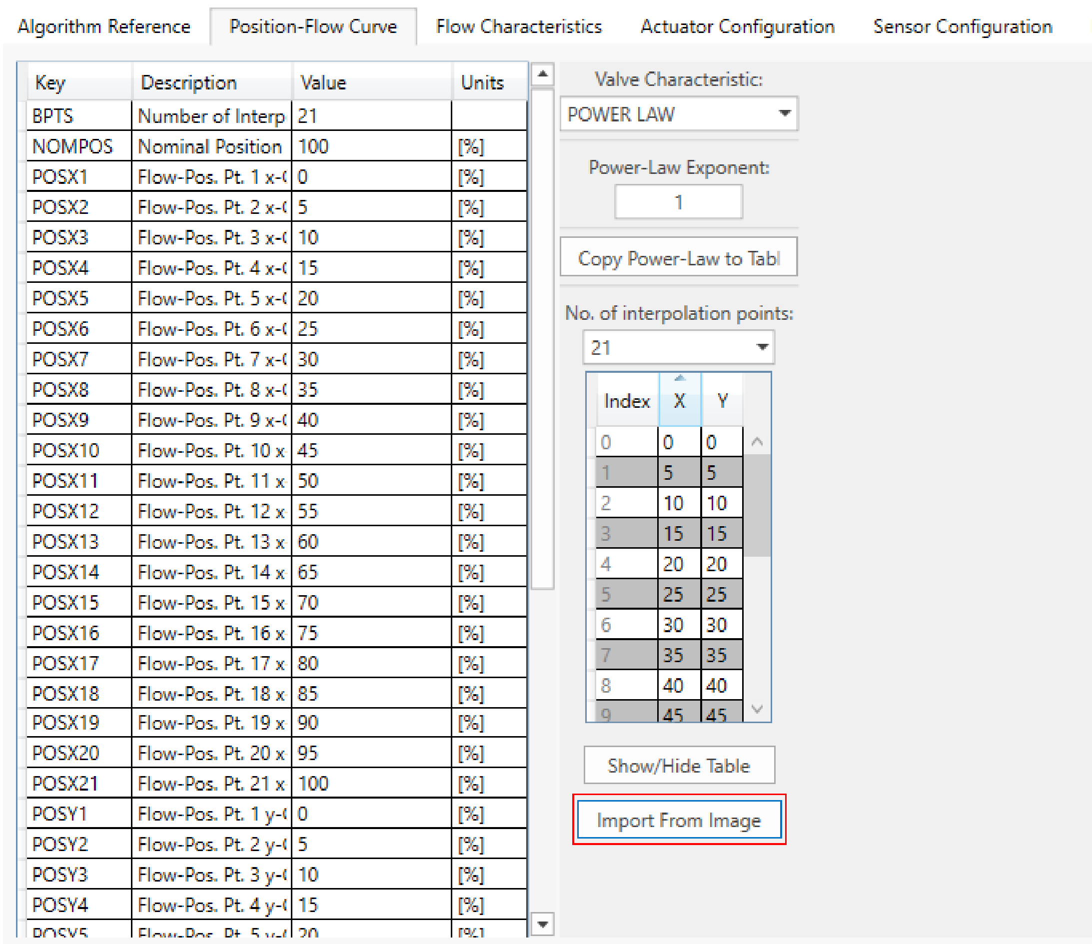
- **Image Loading** - Allows you to load various image formats (for example, .jpeg, .png, .bmp) that contain the graph to be digitized.

- **Data Export** - Allows you to export the digitized data in CSV format for further analysis.
- **Zoom-in and Zoom-out** - Allows you to zoom in and out for detailed viewing and precise point selection.
- **Linear Scaling** - Allows digitizing with linear axes.
- **Logarithmic Scaling** - Supports logarithmic scales on either or both axes with appropriate calibration tools.

Use the following steps to digitize the characteristic curve:

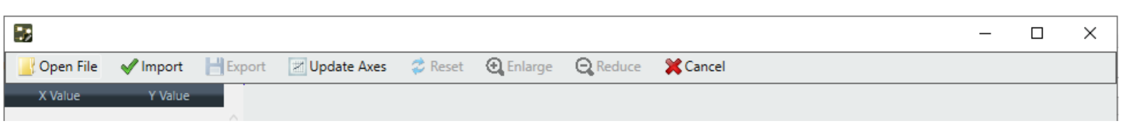
1. Click the **Import From Image** button.

Figure 47. Advanced Edit window- Import From Image button



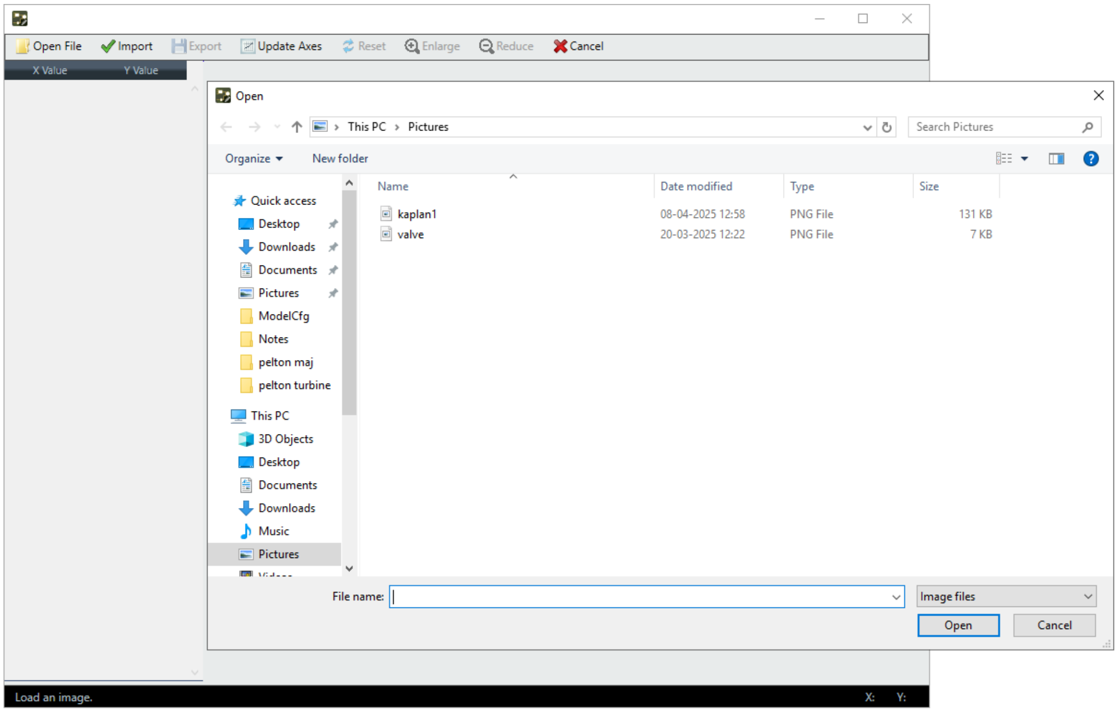
2. The Graph Digitizer appears. To open a graph, click the **Open File** option in the toolbar.

Figure 48. Graph Digitizer window



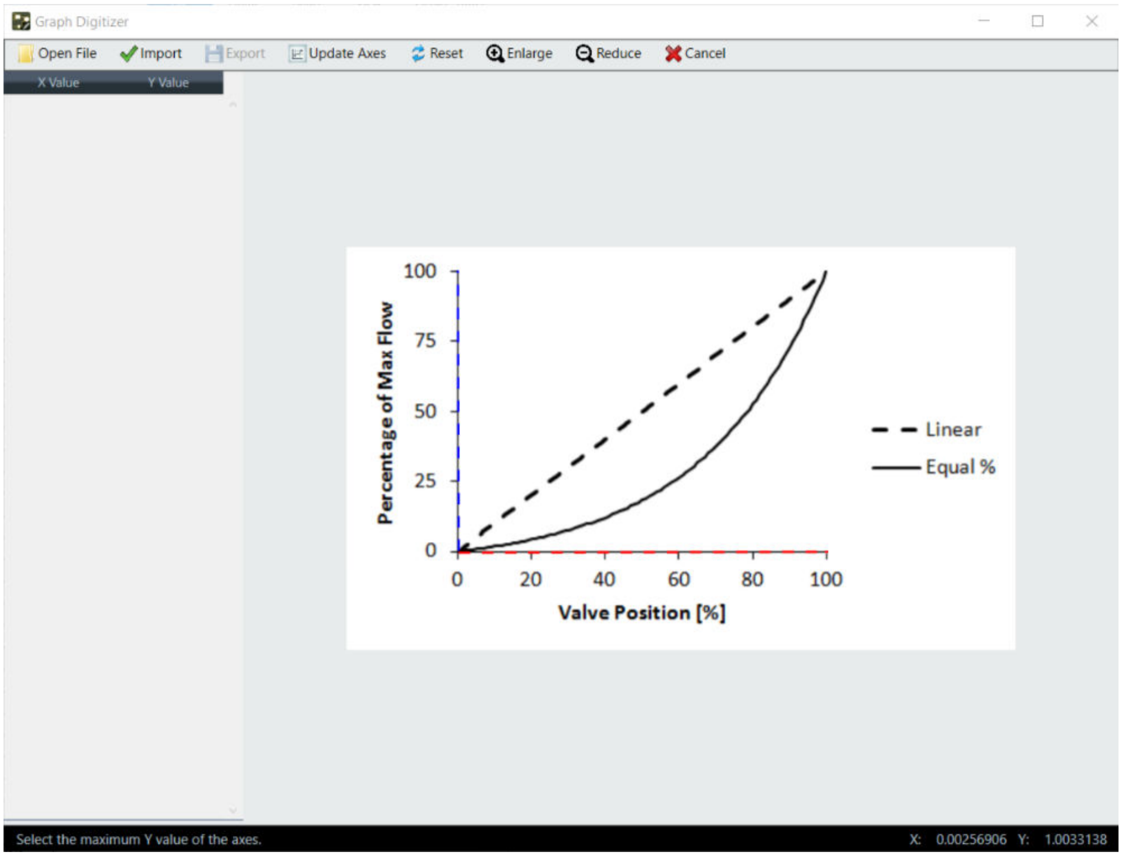
3. From the Open window, select a graph and click **Open**.

Figure 49. Open window



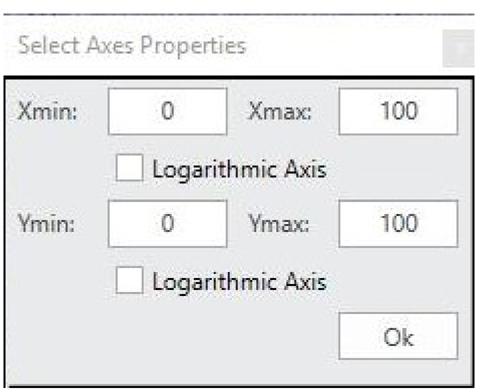
4. From the graph, select the minimum and maximum values of the X and Y axes. The red dashed line on the graph represents the selection made for the X-axis, and the blue dashed line represents the selection made for the Y-axis.

Figure 50. Selected Minimum and Maximum values for both axes



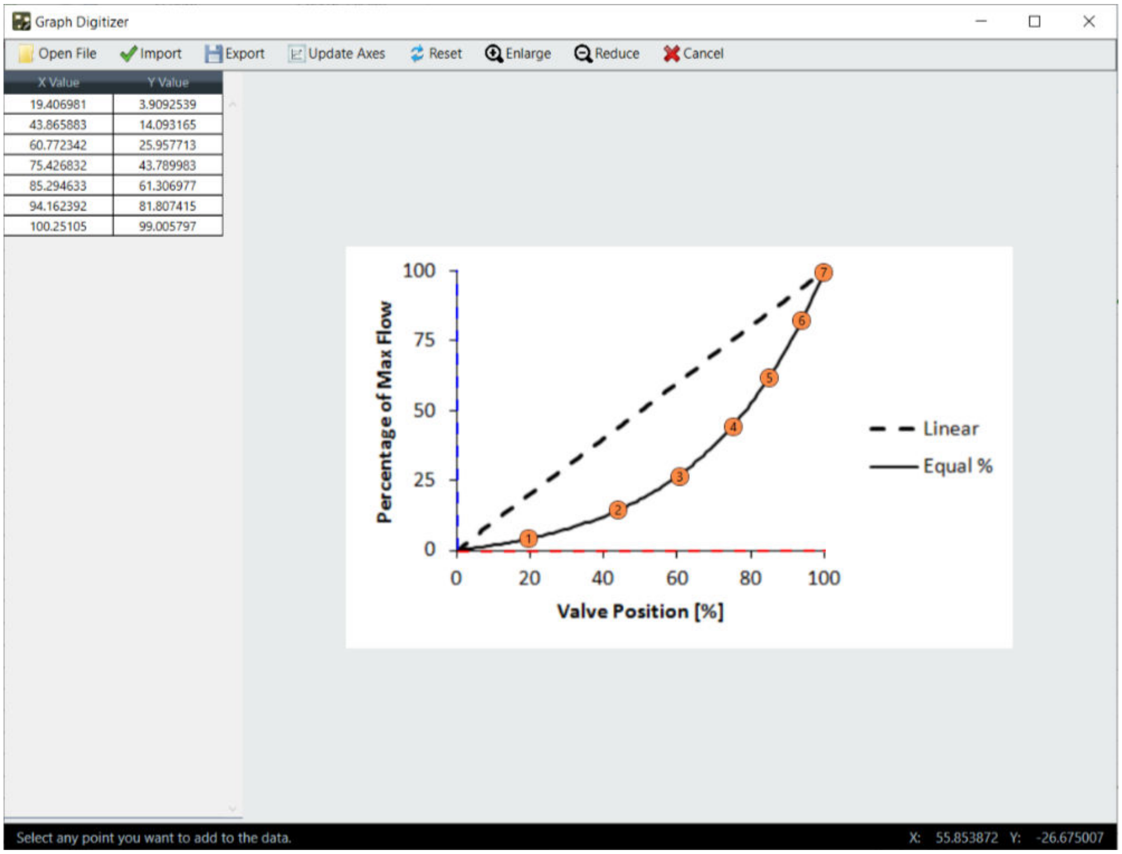
5. A dialog box appears, prompting you to specify the values for each selected axis and providing an option for logarithmic scaling.

Figure 51. Select Axes Properties box



6. Once the graph is fully calibrated, select the points you want to add to the data. The orange numbered indicators on the graph represents the selected points.

Figure 52. Selected Points



7. After selecting all the desired points, click the **Import** option in the toolbar.

After you click the Import option, the Graph Digitizer window closes and automatically transfers the selected point values into the interpolation table within the Advanced Edit window.

5.6.2 Notes Tab

The Advanced Edit tool includes a Notes tab that provides a centralized space to document important information related to the algorithms. The Notes tab allows you to:

- Add detailed notes corresponding to tuning adjustments made to the selected algorithms, store relevant test data, record customer feedback, and maintain a historical log of changes.
- Edit or update existing notes as changes are made.
- Delete obsolete notes when no longer needed.
- View notes for each algorithm.

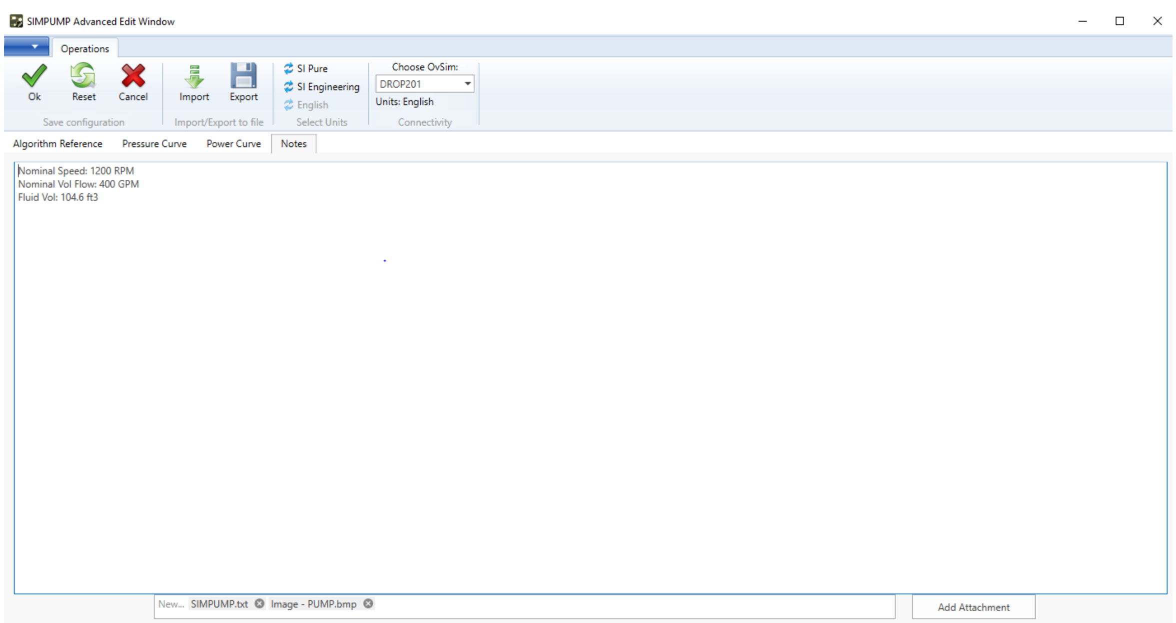
This helps enhance usability, traceability, and collaboration.

To save these notes, click the **OK** button located in the ribbon tab.

CAUTION

- If you rename the point and system ID (SID) of an algorithm, any previously associated notes will no longer be accessible.
- Before making any changes to point names, copy the existing notes for the affected algorithms. After the name change, paste the notes into the Advanced Edit Notes.

Figure 53. Advanced Edit window- Notes Tab



The Add Attachment box located at the bottom right corner of the window allows you to reference digital data files directly within their notes, providing a more comprehensive and integrated approach to documenting tuning adjustments. There are three options to manage digital file attachments:

- **Adding New Attachments** - Adds new attachments by selecting digital files. This supports multiple file selections, allowing you to attach several files at once.

  Note

  - Only the following file types are acceptable for attachments to ensure compatibility and security:
    - **For documents** - .pdf, .txt, .doc, .docx
    - **For spreadsheets** - .xls, .xlsx
    - **For images** - .jpg, .jpeg, .png, .gif, .tiff, .bmp
  - The maximum file size for attachments is 50 MB to manage storage and performance effectively.

- **Opening Attachment** - Opens the attachment when you double-click the name of an attachment.

  Note

  Attachments open in a separate window using suitable third-party applications (for example, Adobe Reader for PDFs, Notepad for text files) are acceptable.

- **Removing Attachment** - Removes the attachment when you click the remove (x) button next to the file name.

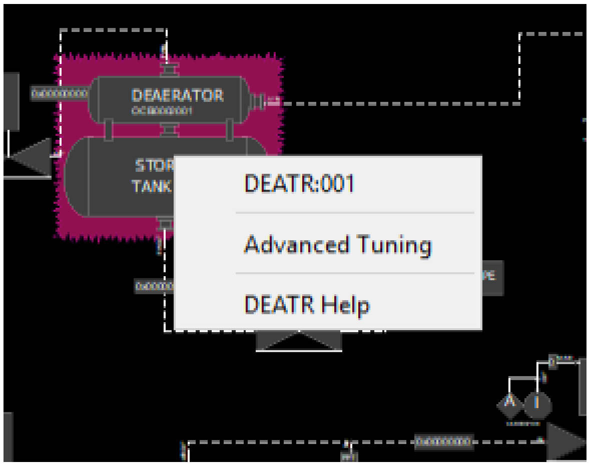
5.7 Ovation Digital Twin Advanced Tuning

The Ovation Digital Twin Advanced Tuning Interface (ATI) provides a specialized, algorithm-specific view of the algorithm states, beyond the simple output parameters of Signal Diagram. For some algorithms, the ATI also provides a means to adjust select tuning parameters and/or state variables.

ATI provides an advanced look into the runtime conditions of a particular algorithm, as well as the underlying structure of that algorithm. It displays the networks associated with the selected algorithm including flow, electrical, and mechanical components, and allows you to view network diagrams along with the medium properties of the elements used. When an element is selected, its runtime properties appear.

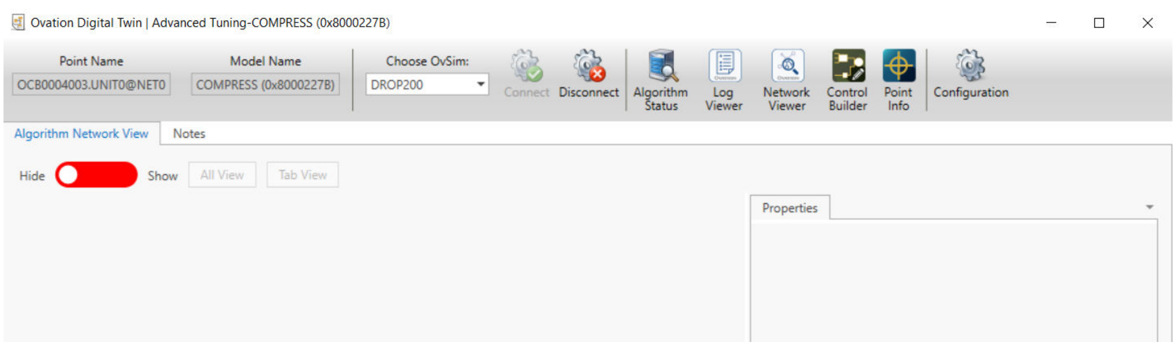
To open the Ovation Digital Twin Advanced Tuning tool, right-click the algorithm in the Ovation Signal Diagram and select **Advanced Tuning** from the popup menu.

Figure 54. Advanced Tuning menu option in DEATR algorithm point



The Advanced Tuning window for the algorithm opens.

Figure 55. Advanced Tuning window - Toggle button set to Hide



A ribbon tab located at the top of the window always loads first. It provides the following functions:

- **Point Name** - Name of the Ovation point attached to the algorithm.
- **Model Name** - Name of the model.
- **Choose OvSim** - Chooses the simulation instance to connect to. Usually, only one simulation instance is in a system.
- **Connect/Disconnect** - Connects and disconnects the simulation.
- **Applications** - Allows you to navigate to the Ovation Digital Twin Algorithm Status, Ovation Digital Twin Log Viewer, Network Viewer, Control Builder, and Point Info applications.

Note

Ovation Digital Twin Algorithm Status, Ovation Digital Twin Log Viewer, and Ovation Digital Twin Network Viewer are included in the Ovation Digital Twin installation and are discussed in this manual. For information on using Control Builder or Point Info, refer to the *Ovation Control Builder User Guide*, *Ovation Operator Station User Guide*, or contact your Emerson representative.

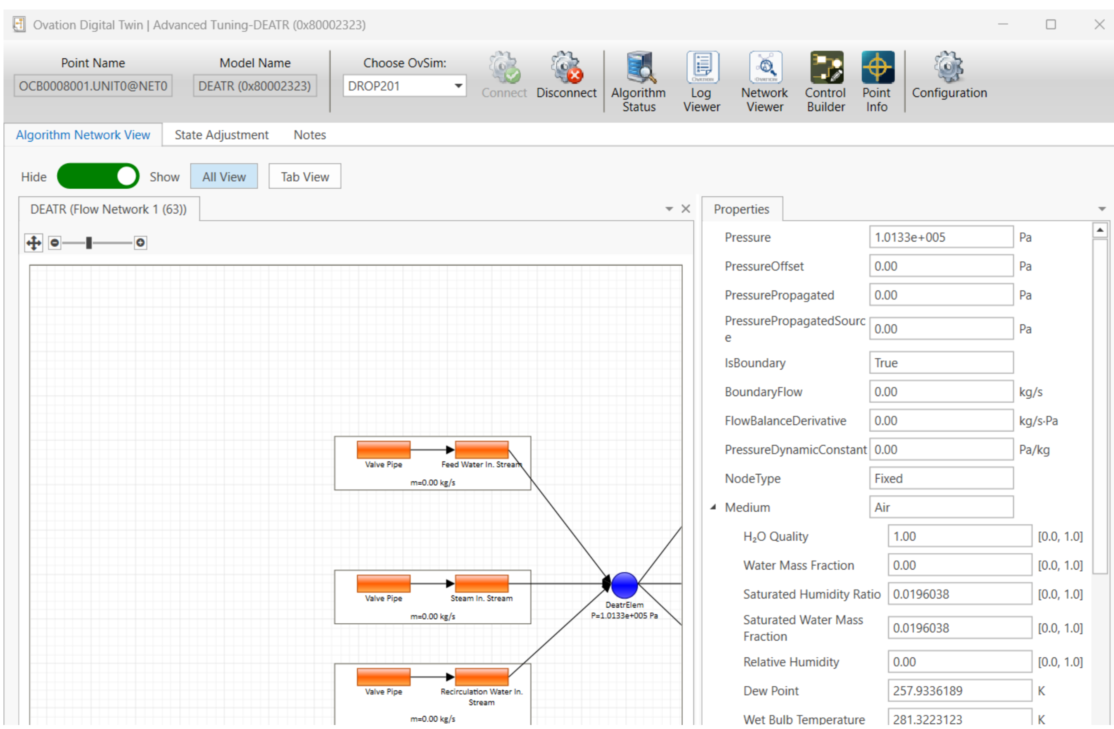
- **Configuration** - Allows you to change the Display Units, Update Frequency, and Tree Depth. You can also specify which data should be displayed at the nodes, from Algorithm Information to Medium Properties (AlgorithmName, DropNumber, Density, Enthalpy, and so forth). For more information, see *Configuration window* (page 57).

The Algorithm Network View section is displayed based on the toggle control. You can control the visibility of the Algorithm Network View using the Hide/Show toggle button at the top-left of the window.

- **Hide (Red Toggle)** - When the toggle is set to Hide, the Algorithm Network View is hidden, and the right pane is cleared. See *Advanced Tuning window - Toggle button set to Hide* (page 66).
- **Show (Green Toggle)** - When the toggle is set to Show, the Algorithm Network View appears. It includes the following:
  - Network View (depending on the algorithm, this can represent a flow, electrical, or mechanical network)
  - Corresponding Properties pane on the right side

  The Algorithm Network View is similar to the Network Viewer interface. For more information, see *Ovation Digital Twin Network Viewer* (page 53).

Figure 56. Advanced Tuning window - Toggle button set to Show



Note

- You can change the toggle state anytime during the session without reconnecting or reloading the ATI.
- The ability to toggle different views helps focus on different tasks by decluttering the Interface when the network view is not required. Emerson recommends that you keep the Algorithm Network View hidden unless you are actively working with it. Keeping the view hidden helps optimize overall performance, especially during extended tuning sessions.

The Algorithm Network View has two options:

- **All View** - Displays all the available networks for the selected algorithm on the same screen.
- **Tab View** - Displays all the available networks for the selected algorithm in a tabbed format. You can then view them by selecting that network tab.

The Algorithm Network View is divided vertically into two panes:

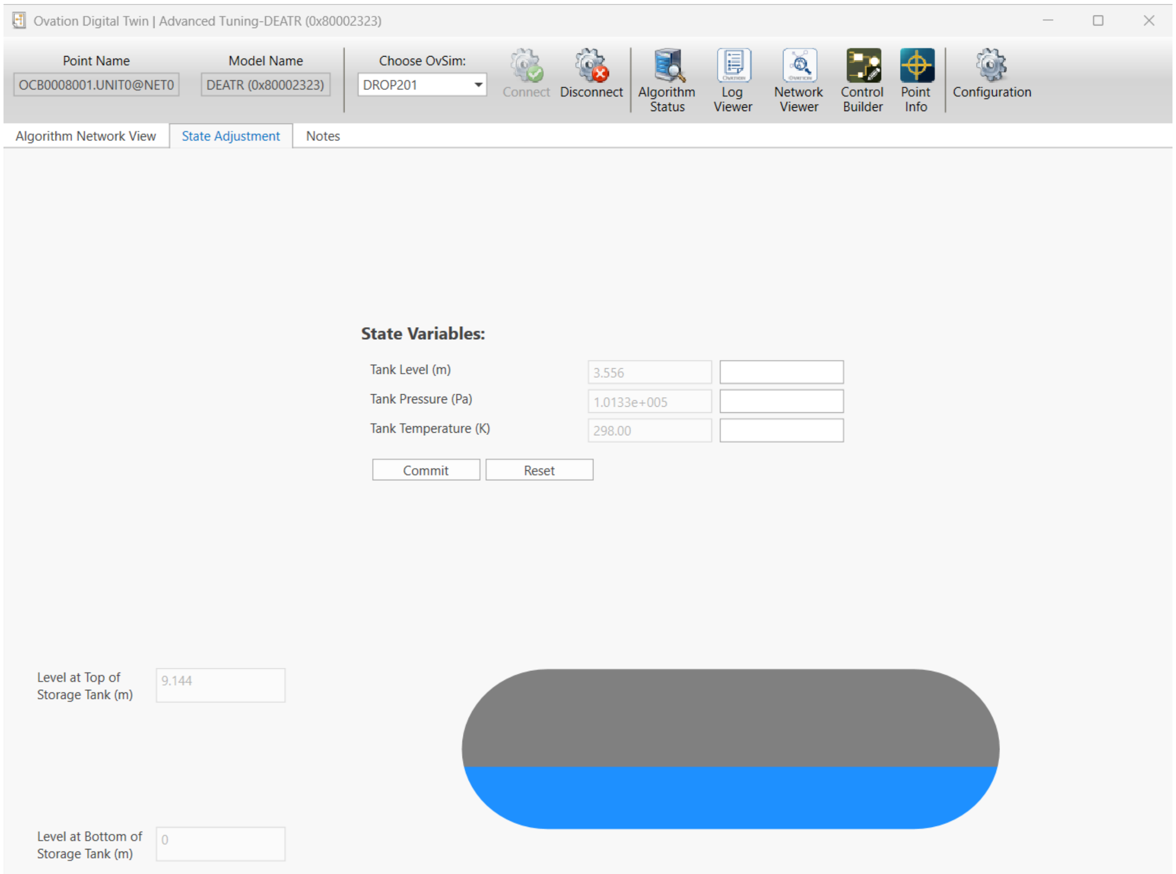
- **Network diagram (left pane)** - Displays the network diagram for the network you selected.
- **Properties (right pane)** - Displays the properties of the selected algorithm.

Depending on the algorithm, additional tabs may be provided to help you with the testing/tuning. One example is a tab for “State Adjustment,” which allows you to adjust the level in the DEATR algorithm.

Note

Since these functions are algorithm-dependent, they are described in the algorithm reference documentation. Contact your Emerson representative for the applicable algorithm reference documents.

Figure 57. State Adjustment Tab



Additionally, the Notes tab allows you to view notes and digital attachments for each algorithm.

5.8 Ovation Digital Twin Media Tables

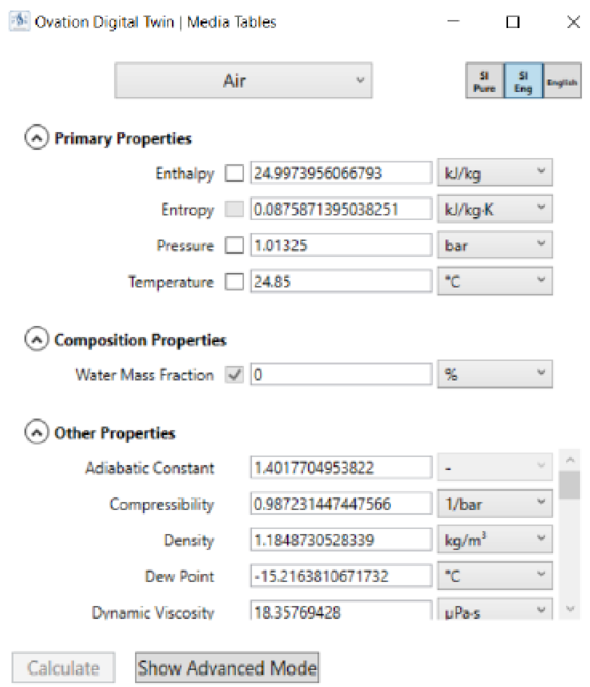
The Ovation Digital Twin Media Tables application allows you to use Ovation Digital Twin built-in property tables for all media supported by Ovation Digital Twin.

To access Ovation Digital Twin Media Tables, double-click the **Ovation Digital Twin Media Tables** icon on your desktop or navigate to C:\Ovation\OvSim\OvSim.MediaTables.exe.

Figure 58. Ovation Digital Twin Media Tables icon

The Ovation Digital Twin Media Tables window opens.

Figure 59. Ovation Digital Twin Media Tables window



The options available vary based on the selected medium. Each medium is characterized by three types of properties: Primary, Composition, and Other Properties.

Fill in the Primary and Composition Properties as appropriate and click the **Calculate** button (see *Ovation Digital Twin Media Tables window* (page 70)).

Note

A minimum of two primary properties are needed for the calculation.

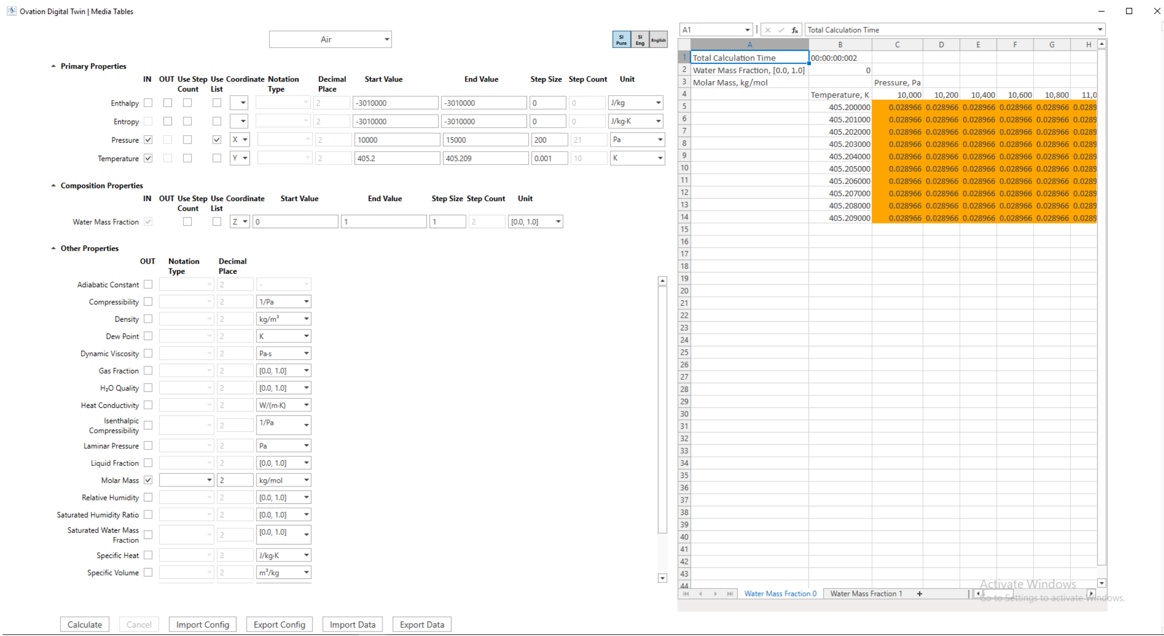
Recalculated properties for the medium conditions you have selected should appear.

5.8.1 Advanced Mode

To access the additional features and options within the interface, click the **Show Advanced Mode** button (see *Ovation Digital Twin Media Tables window*).

The Advanced Mode window opens.

Figure 60. Ovation Digital Twin Media Tables window- Advanced Mode



The Advanced Mode allows you to perform the same media property calculations as the base mode, except it allows running these calculations on a range of conditions, rather than limited to a single condition point (like the base mode).

The Advanced Mode window provides flexible options to define input ranges and compute multiple output properties with custom formatting.

- **Primary Properties** - These properties can serve as both input and output.
  - **IN/OUT Checkboxes** - The IN checkbox allows you to use a parameter as an input and the OUT checkbox allows you to compute and display it as an output.
  - **Use Step Count** - When selected, the Step Size text box becomes inactive, while the Step Count text box becomes active, allowing you to configure the total step count. However, if the Use Step Count checkbox is not selected, the Step Count text box is grayed out, and the Step Size text box becomes active. This allows you to configure the step increment number.
  - **Coordinate Dropdown** - Allows you to assign the input parameter to the X, Y, or Z axis.
  - **Notation Type** - Allows you to choose between float or exponential for results formatting. The Notation Type checkbox is grayed out if the IN checkbox is selected.
  - **Decimal Place** - Allows you to specify the number of decimal places to display in the output. The Decimal Place checkbox is grayed out if the IN checkbox is selected.
  - **Start Value and End Value** - Allows you to define a range for the primary properties. If the start value is greater than the end value for any input parameters with a negative step increment, the system displays the negative increment calculation.
  - **Step Size and Step Count** - Allows you to configure the step size and step count.
- **Composition Properties** - These properties can only serve as inputs.
- **Other Properties** - These properties can only serve as outputs.

Specific limitations exist for specifying both range and static combinations. The available options include:

- You can choose a single range parameter while keeping the other parameters static. The results appear in a row format, similar to an Excel spreadsheet.
- You can choose two range parameters while keeping the other parameters static. The results appear in a tabular format, similar to an Excel table.
- You can choose three range parameters while keeping the other parameters static. The results appear on separate sheets, each containing tables of results.

At the bottom of the window, there are several buttons that allow you to perform different actions:

Figure 61. Ovation Digital Twin Media Tables window- Feature Buttons

- **Calculate Button** - Allows you to perform computations on the media tables. You can specify a range for input parameters and indicate which parameter should undergo calculation by utilizing the OUT checkbox. The media table tool processes the results and presents them in a spreadsheet control.
- **Cancel Button** - Allows you to cancel ongoing calculations on the media tables. This button is available while calculations are in progress.
- **Import Config Button** - Allows you to import the entire configuration of the media table, including the range and media properties values for each property of the media. With this feature, you can choose a configuration file, and all setups will be configured according to that file.
- **Export Config Button** - Allows you to save the configuration of the media table, encompassing the range and media property values for each property.
- **Import Data Button** - Allows you to import the results of configurations configured by you. You can choose a data file, and the data will be displayed in the spreadsheet control. Support is provided for csv format or xlsx format including color effect and visualization.
- **Export Data Button** - Allows you to save the computed data results from the media table tool for configurations configured by you. Support is provided for csv format or xlsx format including color effect and visualization.

Note

There is no provision to delete configuration and data result files from the media table application. You have the option to delete these files according to your needs since the files are not stored in a specific location. You can save them in a location of your choice and can delete them if needed.

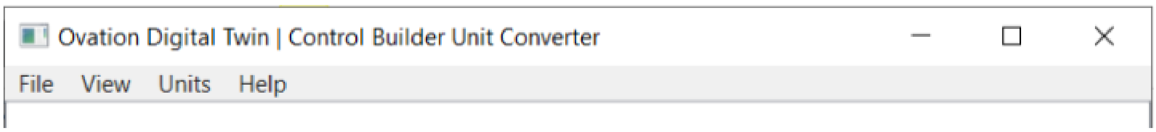
5.9 Ovation Digital Twin Control Builder Unit Converter

The Ovation Digital Twin Control Builder Unit Converter provides a specialized interface for converting engineering units used in control algorithms within the Ovation Digital Twin environment. It provides a way to reuse Ovation Digital Twin control sheets between systems using different engineering units. The Ovation Digital Twin Control Builder Unit Converter supports adaptation of control sheets for various regional and project-specific standards.

To access the Ovation Digital Twin Control Builder Unit Converter, navigate to C:\Ovation\OvSim\OvSim.CbUnitConverter.exe.

The Ovation Digital Twin Control Builder Unit Converter window opens.

Figure 62. Ovation Digital Twin Control Builder Unit Converter window



A ribbon tab located along the top of the window provides access to the following functions:

Table 2. Ribbon tab functions

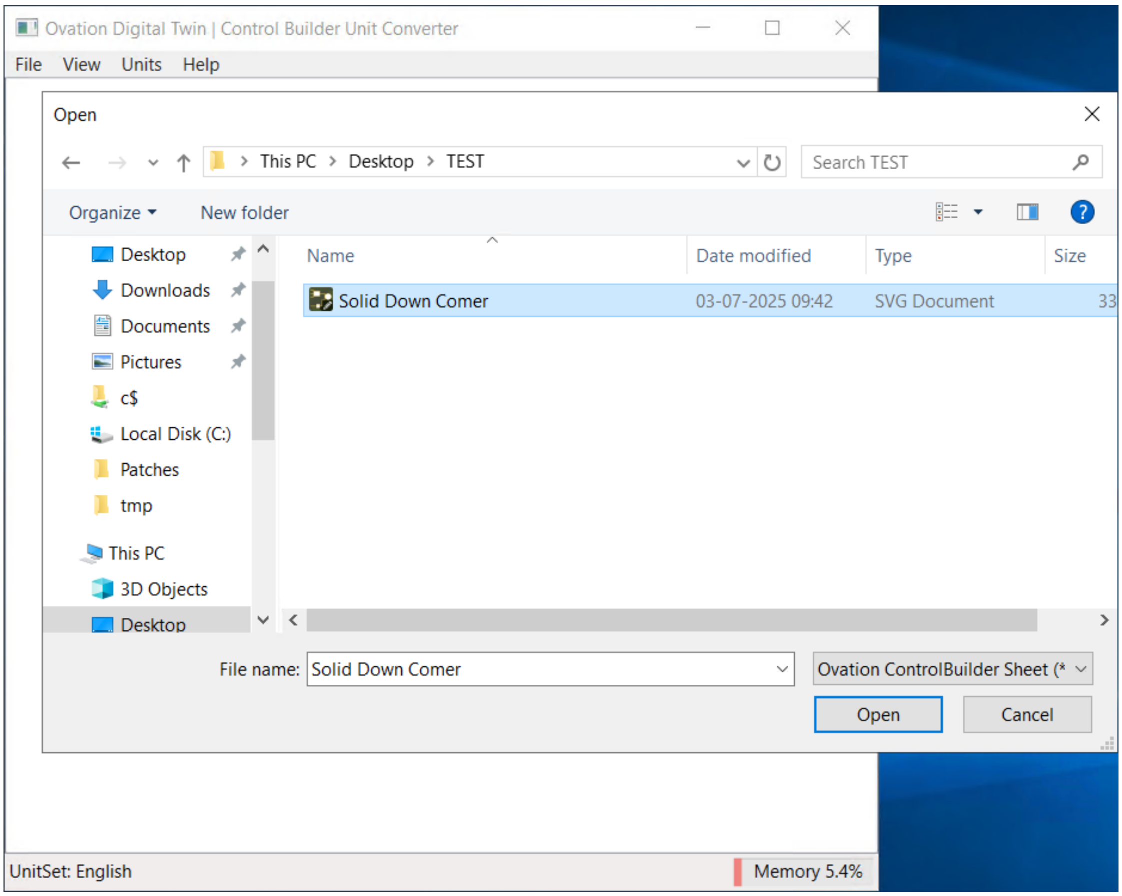
<table>
<tr><th>Tab</th><th>Options</th><th>Description</th></tr>
<tr><td rowspan="5">File</td><td>Open Sheets</td><td>Loads the control sheet. You can load one or more control sheets at a time.</td></tr>
<tr><td>Convert Selected</td><td>Converts the units for selected algorithms of a control sheet.
Note
Do not perform any conversions until you have configured the units in the Units menu.</td></tr>
<tr><td>Save Results</td><td>Saves the converted data.</td></tr>
<tr><td>Clear Sheets</td><td>Clears all data from the view.</td></tr>
<tr><td>Close</td><td>Exits the tool.</td></tr>
<tr><td>View</td><td>Conversion Log</td><td>Displays a list of all previous conversions.</td></tr>
<tr><td rowspan="3">Units</td><td>Current Units</td><td>Selects the current engineering unit used in the control sheet. This is the unit of the simulator from which you are importing the control sheet.</td></tr>
<tr><td>Target Units</td><td>Selects the engineering unit you want to convert to. This should match the Ovation Digital Twin Solver Unit Set of the simulator that will receive the imported control sheet (see Ovation Digital Twin Config (page 44) for information on TrayApp).</td></tr>
<tr><td colspan="2">Note
You must select both the current and target units before performing a unit conversion.
The available unit systems are SiPure, SiEngineer, and English. Emerson does not recommend using the SiPure system.</td></tr>
<tr><td>Help</td><td>Known Algorithms</td><td>Displays a list of all Ovation Digital Twin algorithms.</td></tr>
</table>

5.9.1 To convert units

Use the following steps to perform the unit system conversion:

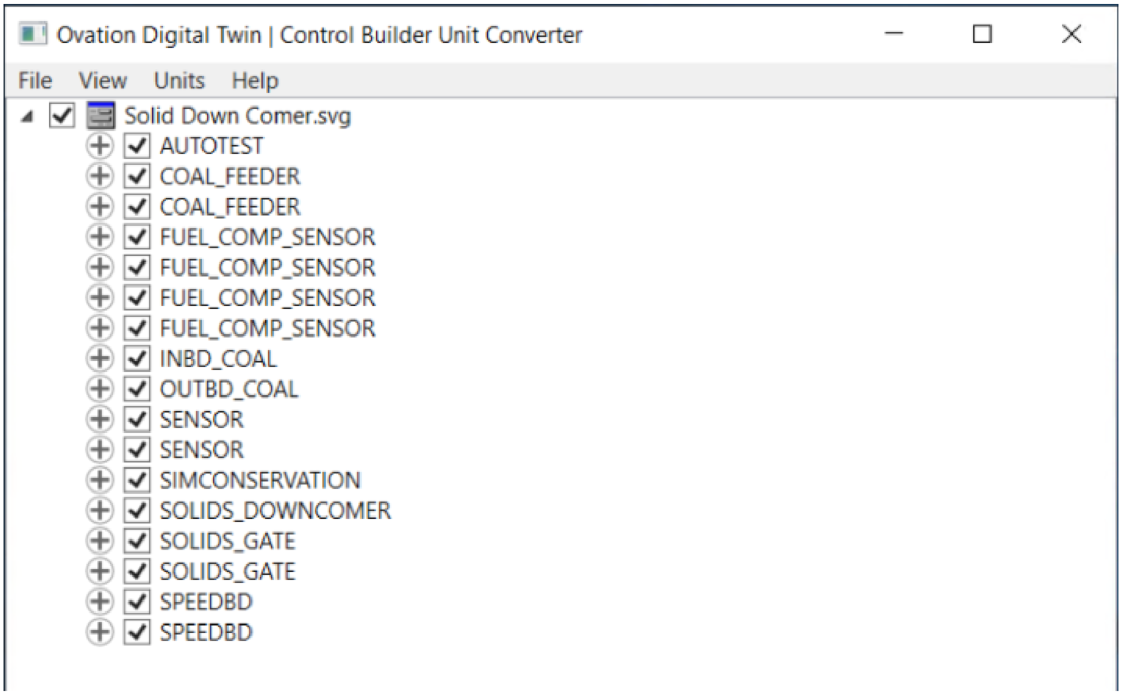
1. Access the Ovation Digital Twin Control Builder Unit Converter.
2. Click **File** > **Open Sheets** in the tool bar.
3. From the Open window, select a control sheet and click **Open**.

Figure 63. Open window



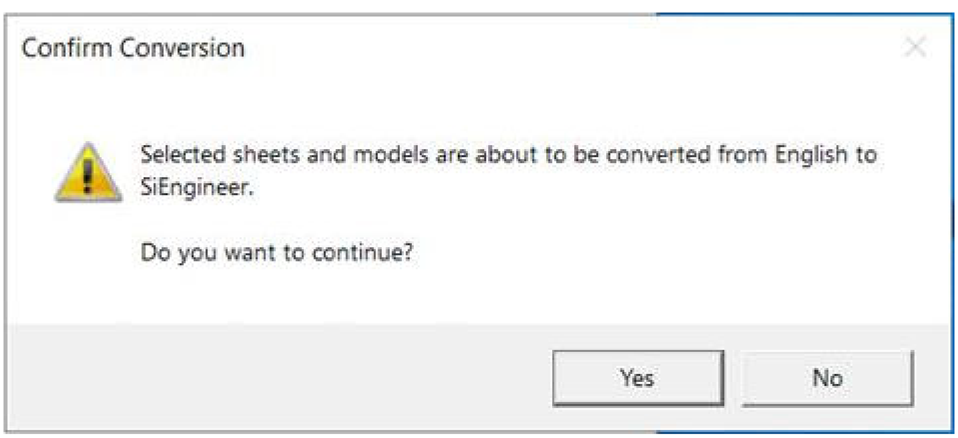
4. The control sheet name appears at the top with a dropdown menu to view its algorithms. Each algorithm has a checkbox to select it for conversion and a + button to view its parameters.

Figure 64. Control Sheet loaded



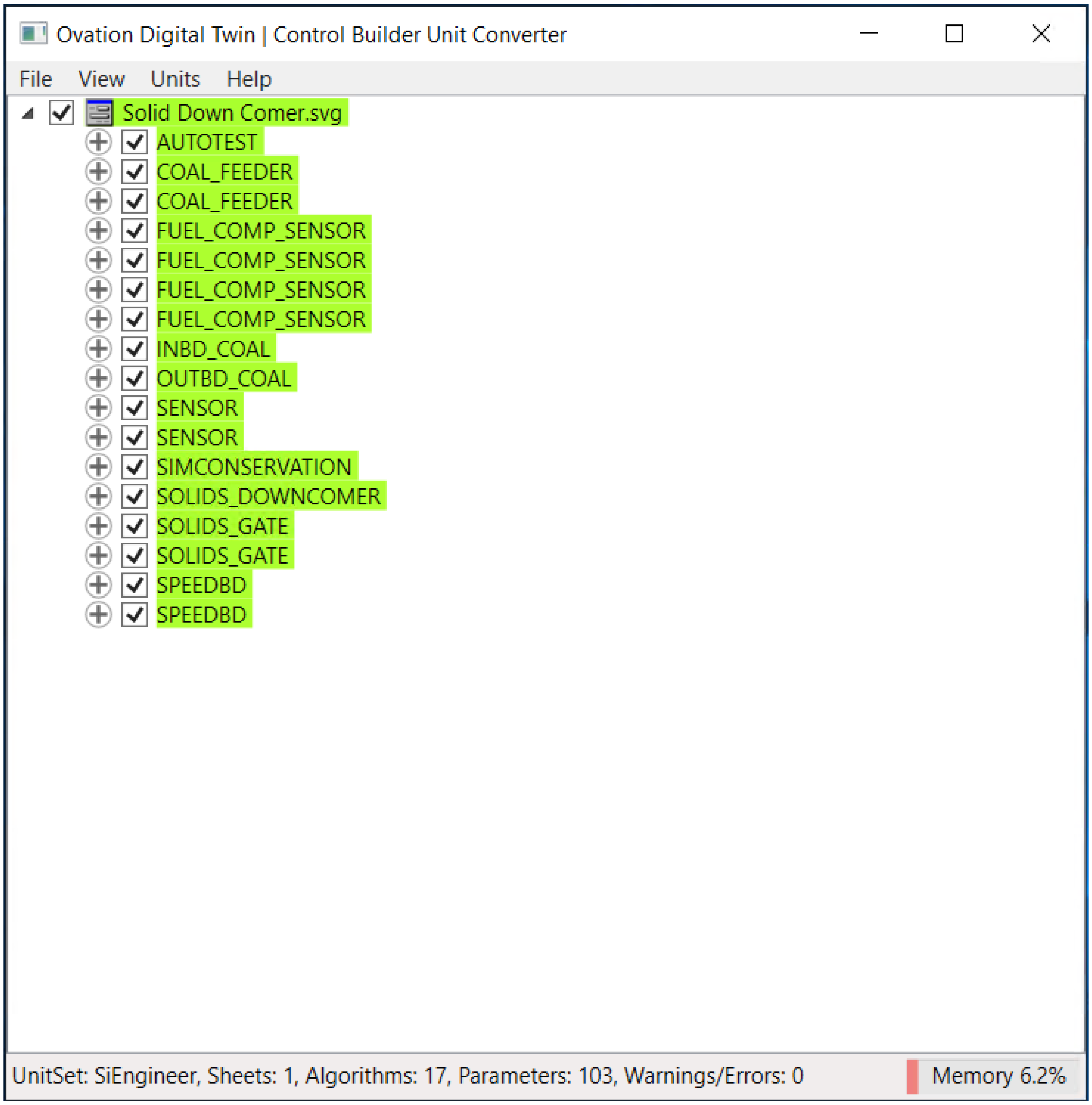
5. Click **Units** > **Current Units** and select the current unit.
6. Click **Units** > **Target Units** and select the unit you want to convert to.
7. Click **File** > **Convert Selected** to perform the conversion.
8. A Confirm Conversion box appears, prompting you to confirm whether you want to convert the selected sheets and models from Current Units to Target Units.
   Click **Yes** to continue. The converted algorithms are highlighted in green.

Figure 65. Confirm Conversion box



9. Click the **+** button next to each green highlighted algorithm to review the converted parameters and ensure values are correct. The status bar at the bottom of the window provides a quick summary of the current unit set, number of sheets, algorithms, parameters, and any warnings or errors.

Figure 66. Converted Algorithms in green color



10. To save the converted control sheet, click **File** > **Save Results**.

Note

Save the converted control sheet with a new name (for example, Control_Sheet_SiEngineer) to retain the original version. Emerson recommends that you include the units in the name to avoid confusion.

The converted control sheet is now ready for import into the database.

6 Troubleshooting

Topics covered in this section:

6.1 Performance Troubleshooting

The following topics provide an overview of performance troubleshooting the Ovation Digital Twin. Refer to the specific topics for detailed instructions:

- *Network Time Protocol (NTP) (page 77)*.
- *Degree of Parallelism* (page 77).
- *To delete the simalgs.cfg from the DCS controller drops* (page 79).

6.1.1 Network Time Protocol (NTP)

NTP identifies the drop or external server that is providing the time base for the Ovation system.

CAUTION

- It is required to have the NTP configured on the Ovation Digital Twin.
- To configure the NTP, refer to the *Ovation Developer Studio User Guide*.

6.1.2 Degree of Parallelism

The Degree of Parallelism is an Ovation Digital Twin configuration parameter that defines the degree at which the simulation calculations (controller communication, network and models calculations) are performed in parallel threads. This parameter is located in the Configuration tab of the Global parameters configuration window (page 44).

What is the role of the Degree of Parallelism parameter?

Different Degree of Parallelism can give different loop time calculation times as ~200 ms for Degree of Parallelism equal 2 (which is too small for all but the smallest for demonstration or test purpose simulator systems) and ~100 ms for a properly configured system.

The configuration of Degree of Parallelism parameter depends on the available model server CPU threads number. It can also be affected by the following factors:

- If the model server is a separate hardware machine or a virtual machine.
- If the number and sizes of large flow networks.
- If the number of simulation algorithms.
- If the model server is also a VCH or an Instructor Station machine.

To configure the Degree of Parallelism

Perform the following procedures to configure the Degree of Parallelism:

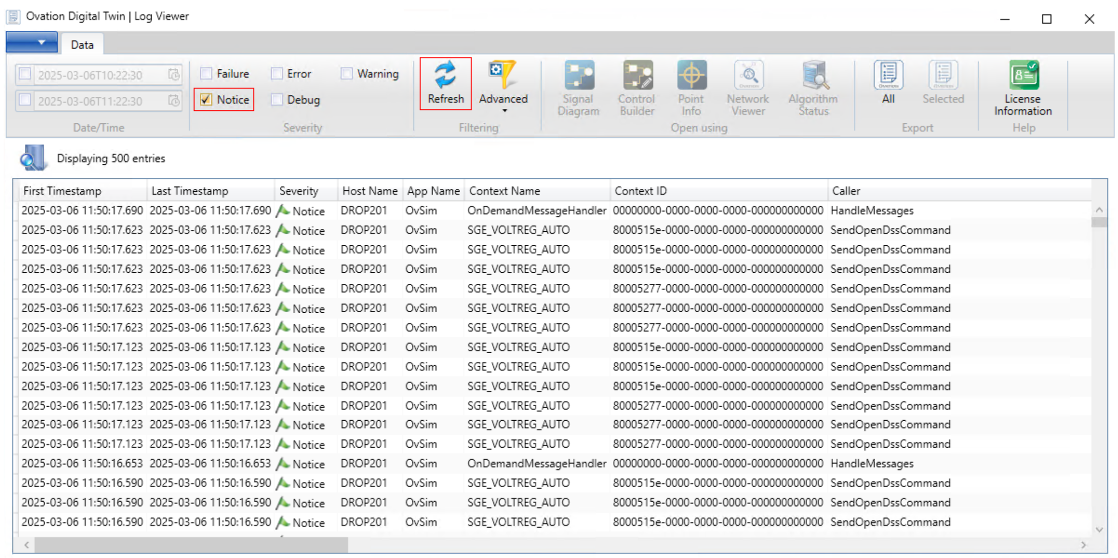
1. Load a reference steady condition Snapshot.

Note

Should not be a cold condition.

2. Access the Ovation Digital Twin LogViewer (page 52). Perform the following step:
   a. Click **Refresh**.
   b. Select **Notices** check box.
   c. Check the average Loop time displays in the window.

Figure 67. Ovation Digital Twin LogViewer



3. Access the Ovation Digital Twin Service Config (page 44). Perform the following steps:
   a. Select **Performance** tab.
   b. Set the initial Degree of Parallelism equal to the number of processor cores minus 2.
   c. Click **Reconcile**.
   d. Click **Apply**.
4. Return to the Ovation Digital Twin LogViewer. Allow the simulation to run for a few minutes and record a few average loop times statistics per value of Degree of Parallelism.
5. Repeat Step 4 with increased or decreased values of Degree of Parallelism (for example. by two, or four) until a significant increase in the average loop time.

Note

Depending on your system, this process may take significant time.

6. Choose the optimal value from the tested configurations.

Note

On some systems with large CPU requirements, the optimal value can exceed 16.

6.1.3 To delete the simalgs.cfg from the DCS controller drops

After installing Ovation Digital Twin on a Virtual Controller Host (VCH) where DCS Virtual Controllers are located, you must delete simalgs.cfg from each DCS drop folder (drop where no simulation models are located). Deleting simalgs.cfg increases network performance between the controllers and Ovation Digital Twin.

The simalgs.cfg file is located at the following path:

C:\Ovation\dropX\ata0\wdpf\dpu\config\simalgs.cfg, where X is the number of the drop.

Note

Currently, this process must be performed manually after every Ovation Digital Twin installation.

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